

The active-inference driver model: a reader's handbook

Prepared at Chalmers University of Technology by Jonas Bårgman, with drafting assistance from Claude (Anthropic), August 2026. This handbook grew out of a close reading of the active-inference driver-modeling line — the article, its Supplementary Information, the released code, and the OSF simulation deposit — undertaken to understand the model well enough to build on it. It is shared with the authors of that work in the spirit of a companion text: what a careful outside reader reconstructed, stated plainly enough to be checked. Any errors of reading are ours, and corrections are very welcome.

Version of 28 September 2026: beyond the edition you received (see the notice below).

Revised 3 September 2026. The first edition was prepared on 24 August 2026. Since then we have read the released code more closely, run the model outside its published scenarios, and compared parts of it against human response data for our own purposes, and several passages have been corrected or extended as a result. Every revision is dated in place. The largest are: the planner's elite set and iteration counts (chapter 03); the collision severity factor and the indicator form of the safety-margin term (chapter 07); a line-by-line account of what differs between the three released scenarios (chapter 04); what the belief machinery does when the observation channel is closed (chapters 06 and 08); and two findings from human data about the model's perceptual and preference stages (chapters 03 and 07), carried under a new tag, [Study].

This is a newer version than the one you received. The edition sent to you is dated 3 September 2026. This version, dated 28 September 2026, adds five passages based on work done since then, four of them using naturalistic traffic data (the highD dataset) that we did not have before. Every addition begins with the label "New in this version, 28 September 2026: not in the edition you received" and is set in color in the Word and PDF files. Nothing in the edition you received has been removed. One sentence has been reworded (the reading of the 25 m/s road departures, chapter 5), and the wording you received is quoted beside it. The additions are in:

- chapter 3, "The looming channel, tested against human judgments": the naturalistic check that the earlier text said was missing;
- chapter 5, "Normal driving: the quiet regime": the model in sustained following, now observed rather than extrapolated;
- chapter 5, "What this means for using the model": the response-time relation on real motorway data;
- chapter 5, the 25 m/s road departures: a softer reading;
- chapter 7, Part A: how often the safety-margin and inverse-tau terms act in real traffic.

What this handbook is

This handbook explains the active-inference driver model of Schumann, Engström, Johnson, O'Kelly, Messias, Kober and Zgonnikov (2026, Nature Communications), together with the line it belongs to (Engström et al. 2024; Wei et al. 2024), well enough that a reader can (1) understand how it works, (2) change it deliberately, and (3) know how to validate a change. It is written for a mixed audience of traffic-safety analysts and human-factors researchers. The main text of every chapter avoids mathematics; each chapter ends with layered notes for readers who want the equations.

The handbook is grounded in five kinds of sources, and every substantive claim is tagged with its provenance:

- **[Paper]** — the published article
- **[SI]** — its Supplementary Information (where most of the actual definitions live)
- **[Code]** — the authors' released code, which is the ground truth when the paper is ambiguous, and which in a few places defines the model more precisely than the SI does (those places are tagged where they matter)
- **[OSF]** — the authors' own simulation output (the OSF deposit), which lets the handbook show real numbers rather than sketches
- **[Opinion]** — our own readings and judgments, marked so they cannot be mistaken for the authors' claims
- **[Study]** — our own analyses since the first edition: human intervention judgments on video clips of a highway cut-in (2026, unpublished), and the released model run on scenarios outside the published three. Stated so they can be checked; not the authors' claims, and not yet peer reviewed

New in this version: since 28 September, [Study] also covers naturalistic motorway data (the highD dataset: 60 drone recordings of German motorways)

The chapters

Part	Chapter	One line
I	01 Where this comes from	The lineage of the idea, what "free energy" actually means, how it relates to models you already know, and the debate around it
	02 One event through the model's eyes	A single rear-end conflict, told moment by moment with real numbers
II	03 What the model is	Every component, its inputs and outputs, and the loop that connects them
	04 Scenario playbook	Exactly what changes between scenarios, and the checklist for adding a new one
	05 Normal versus critical	Why the same machinery covers everyday driving and emergencies
	06 Other agents and beliefs	What the other vehicle actually does, versus what the driver model believes it might do
	07 Normative driving	What defines "normal" behavior, and every knob that moves it
	08 The gaze system	The attention machinery that ships in the code, and what impairment-related mechanisms exist today
III	09 Modify and validate	Recipes for changing the model, each with its validation ladder
	10 Calibration and parameter fitting	Where every number came from, how to set new ones, identifiability, and the dos and don'ts
IV	11 Code and data map	From concept to file, class, and parameter
	12 Glossary	The same idea in three vocabularies, plus common misconceptions
	13 Appendix: the deep end	The free-energy principle proper, variational inference, Markov blankets, the debate literature, the discrete-state formulation — for reference

Reading paths

- **Thirty minutes, any background:** chapter 02, then the first half of chapter 01.
- **Human-factors readers:** 01 → 02 → 05 → 07 → 08. Chapter 03 on demand.
- **Analytics / modeling readers:** 02 → 03 → 04 → 06 → 09 → 10 → 11, with the chapter-end notes.
- **"I want to change the code":** 03 → 04 → 11 → 09 → 10, keeping 12 open in a second window.

How the math is layered

Each chapter's main text is **Level 0**: components, inputs, outputs, and behavior, no equations. At the end of a chapter:

- **Level 1** states the same content with light notation — enough to read the paper's figures and follow its argument.
- **Level 2** gives the actual equations with their Supplementary Information numbers, so a reader can go from this handbook straight into the SI or the code.

Nothing in a later chapter depends on having read the notes of an earlier one.

One warning before you start

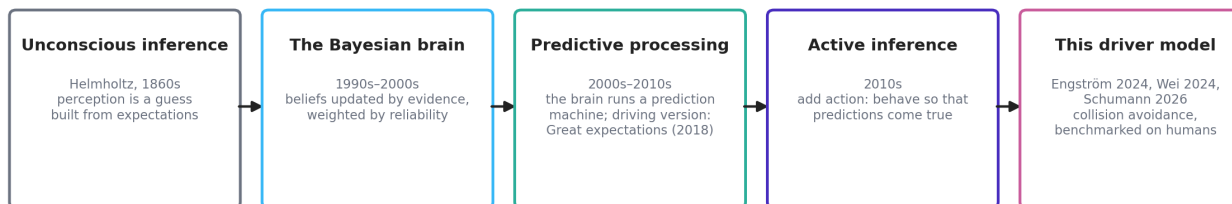
Two words in this literature do not mean what they usually mean, and misreading them is the most common way to get lost. **Surprise** is not an emotion here: it is a number measuring how far what is happening departs from what the model expected. **Preference** is not a choice: it is a description of the futures a driver treats as normal, encoded so that wanting something and expecting it become the same quantity. Chapter 12 collects the rest of these false friends.

Chapter 1: where this comes from

The idea in one paragraph

Active inference starts from a simple claim: the brain is not a camera followed by a calculator. It is a prediction machine. It continuously guesses what its senses are about to report, compares the guess with what actually arrives, and treats the difference — the surprise — as the thing to get rid of. There are only two ways to get rid of it: change your mind (update your beliefs until they fit the world) or change the world (act until it fits your beliefs). The first is

perception. The second is action. Active inference says these are not two systems but one operation running in two directions, and it builds driver models in which detecting a braking lead vehicle and pressing the brake pedal are, literally, the same computation.



A short history, told through what each step added

Unconscious inference (Helmholtz, 1860s). Perception is not passive reception; it is a guess constructed from expectations plus sensory evidence. A century and a half later this is uncontroversial in perception science — the visual system demonstrably fills in, corrects, and predicts.

The Bayesian brain (1990s–2000s). The guessing was given arithmetic: beliefs are held with degrees of confidence, evidence updates them, and less reliable evidence moves them less. The important word is *reliability* — the framework automatically trusts a clear view more than a glimpse in fog, without a separate rule saying so.

Predictive processing (2000s–2010s). The update arithmetic became an architecture: the brain runs a generative model — an internal simulation of how the world produces sensations — and works to minimize prediction error at every level, from retinal input up to "that car is going to cut in". For driving, this is not an imported idea: *Great expectations* (Engström, Bårgman, Nilsson, Seppelt, Markkula, Piccinini and Victor, 2018) laid out a predictive-processing account of automobile driving before the present model line existed. The way we read it, the Schumann model is the computational instantiation of the account that paper gave verbally [Opinion].

Active inference (2010s, Friston and colleagues). Predictive processing explains perception. Active inference adds the second direction: an agent can also reduce prediction error by acting on the world. The trick that makes this work is the **preference prior**: the model treats the futures the agent *wants* as the futures it *expects*. A driver who expects to be traveling safely in their lane, and who finds the world drifting away from that expectation, will act to pull the world back to it. Goal-seeking becomes surprise-avoidance, one currency for both.

This driver model (2024–2026). The Waymo / TU Delft line (Engström et al. 2024, Wei et al. 2024, Schumann et al. 2026) turned the framework into a runnable model of human driving in safety-critical situations: it perceives through optical looming like a human eye, carries realistic uncertainty about what other road users will do, plans a few seconds ahead with bounded effort, and times its responses by accumulating surprise until action becomes necessary. It reproduces human response-time patterns in three conflict types, one of which the authors held out entirely from tuning [Paper].

"Free energy," demystified in one page

The term that scares people off is *free energy*. What it names is mundane: **a computable score of how badly your model of the world is doing**, given what you are sensing. High score, poor fit; low score, good fit. The mathematical object (Level 2 note) is a clever upper bound on surprise that can be computed without knowing everything about the world — that is its entire job.

The word "energy" is a historical accident: the formula has the same shape as a quantity in statistical physics, so the name was borrowed. Nothing thermodynamic is meant. No heat, no calories, no metabolic claim. A reader who mentally substitutes "model-misfit score" for "free energy" loses nothing in this handbook and, in our experience, most of the model literature [Opinion].

Two versions of the score matter, and the split between them organizes the whole model:

- **Present-tense misfit** (variational free energy): how badly current beliefs fit current sensations. Minimizing it is perception — the belief update.

- **Future-tense misfit** (expected free energy): how badly a *candidate plan* is expected to fit the *preferred* future. Minimizing it is action selection — choose the plan whose imagined consequences least depart from the future the driver treats as normal.

Anchors: models you already know, and where they sit inside this one

The model is best understood not as a rival to the models this audience already uses but as a container that holds versions of them [Paper] [SI]:

- **Evidence-accumulation / drift-diffusion response models** (Markkula and colleagues). The model's response timing *is* an accumulator: a quantity builds up over time and action follows when it crosses a threshold. The difference is that the accumulation rate is not a fitted constant — it is the moment-by-moment surprise computed from the driver's own predictions, so response times automatically depend on kinematics, urgency, and expectation.
- **Looming and visual threshold models**. The model perceives the lead vehicle through optical size and expansion rate, with a detection threshold on expansion. Detection delay therefore *emerges* from perception rather than being a fitted reaction-time constant.
- **Driver risk field and safety-margin models** (Kolekar and colleagues). The preference prior defines a landscape over states — which situations are treated as normal and which as increasingly unacceptable. That landscape plays the role of a risk field.
- **Motivational theories** (zero-risk, task-capability interface, task difficulty homeostasis). These describe drivers as regulating toward a comfortable region. Here the regulation is explicit: the comfortable region is where predicted futures match preferred ones and surprise stays near zero.

The debate: biology, philosophy, or engineering

Active inference arrives with a large and sometimes heated literature attached, and it is fair for a new reader to ask what they are being asked to believe. The way we read the field, three distinct claims travel under the same name, and they deserve different levels of commitment [Opinion]:

- 1 **A process theory of the brain** — neurons literally implement these computations. This is a serious neuroscience program with real but contested evidence. Nothing in this handbook depends on it.
- 2 **A universal principle of life** — every self-organizing system, from cells upward, minimizes free energy; in its strongest form this is argued almost as a mathematical necessity. Critics respond that a principle compatible with everything predicts nothing, and that the strongest versions are unfalsifiable. We do not take a position here, and we do not need to.
- 3 **An engineering framework** — a recipe for building agents that carry uncertainty honestly, unify goal-seeking with information-seeking, and time their actions by surprise. This is the only claim the driver model needs, and it is testable the ordinary way: build the model, benchmark it against human data, hold scenarios out.

The published model is squarely a use of claim 3, and its strongest evidential card is conventional science rather than grand theory: the intersection scenario was never used for tuning, and the model still predicted human response patterns there [Paper]. That said, the framework's critics raise points worth keeping in view even at level 3: with a hand-built preference function and thirteen tuned parameters, flexibility is real, and "the model can be made to fit" is a fair worry. The honest defense is held-out prediction and ablation — remove a mechanism, show the behavior degrades — both of which the paper does, and both of which any reader can now reproduce from the released data [OSF].

For contrast with the paradigms this audience grew up with:

Paradigm	The driver is...	Where it differs from active inference
Stimulus–response / threshold models	a trigger waiting for a cue	no expectations; response times must be fitted per situation
Information processing (perceive-decide-act stages)	a pipeline	stages are separate boxes; here perception, decision, and timing share one currency
Ecological psychology (Gibson)	attuned to optical invariants	closer than it looks — looming <i>is</i> an optical invariant; active inference adds explicit beliefs and preferences behind the optics

Paradigm	The driver is...	Where it differs from active inference
Optimal control	a perfect planner with a cost function	here planning effort is deliberately bounded, and the "cost function" is a probability distribution, which is what lets surprise time the response
Reinforcement learning	a reward maximizer trained by experience	no reward signal exists here; preferences are built in, not learned, and information-seeking comes for free rather than needing exploration bonuses

Readers who want the deep end — the free-energy principle proper, variational inference, Markov blankets, and the debate literature on both sides — will find it in chapter 13, which exists precisely so that this chapter does not have to be longer.

The permission slip

You can use everything in this handbook while remaining entirely agnostic about brains and about universal principles. The model stands or falls on whether it predicts human driving behavior, which is an empirical question with published, partly held-out, answers. That is the spirit in which the rest of the handbook is written.

Notes for the mathematically curious

Level 1 — the two scores. Surprise of an observation is its negative log-probability under the model: rare-according-to-you means surprising-to-you. Variational free energy is an upper bound on surprise that is computable because it involves only the agent's own beliefs and its generative model. Expected free energy scores a candidate policy by the surprise its predicted observations would carry under the preference distribution, plus a term rewarding observations expected to reduce uncertainty. Action selection picks the policy with the lowest score; the two terms are the pragmatic and epistemic values that chapter 03 treats as components.

Level 2 — the objective. With beliefs $Q(s)$, generative model $P(o, s)$, and preference prior $P(o)$:

Perception: minimize over Q : $F = E_Q[\log Q(s) - \log P(o, s)]$ ($\geq -\log P(o)$)
Action: minimize over π : $G(\pi) = -E[\log P(o_{\text{future}})]$ (pragmatic value)
 $-E[\text{information gain}]$ (epistemic value)

The preference prior $P(o)$ is where "wanting" enters: it is a probability distribution over observations in which preferred observations are the probable ones. In the driver model the distribution is a product of six independent terms (speed, acceleration, steering, lane position, closing rate, collision/safety); chapter 07 walks through each. The full forms are SI §2.4, Eqs. 44–52 [SI], and the derivation-level treatment is in chapter 13.

Chapter 2: one event through the model's eyes

Every number in this chapter is read directly from the authors' own published simulation output [OSF] (rear-end scenario, experiment 7, random seed 0) — nothing is sketched or idealized.

The cast

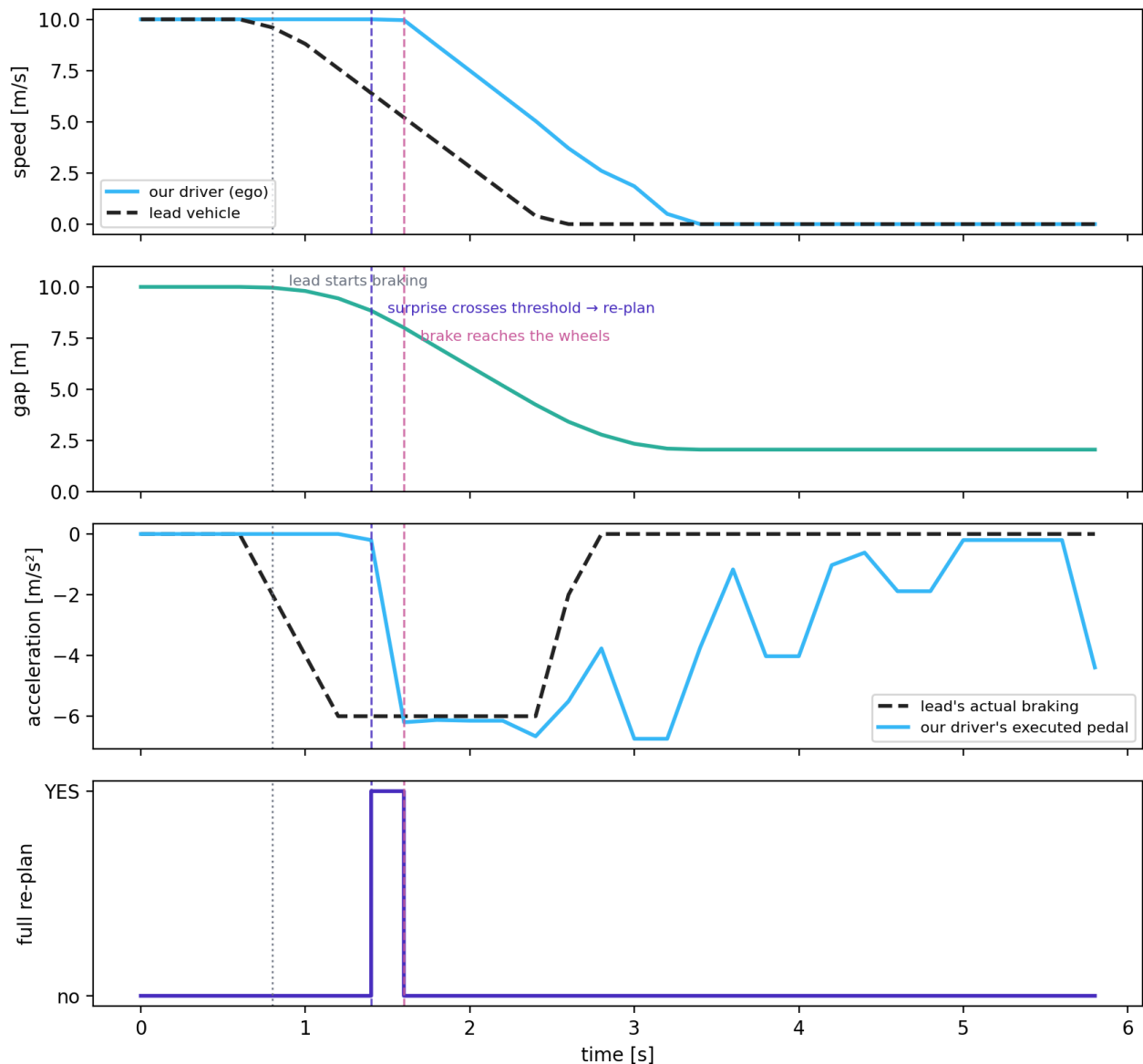
Two vehicles on a straight road. **Our driver** — the model — is following a **lead vehicle**. Both travel at 10 m/s (36 km/h), with a 10 m gap between our front bumper and their rear bumper: exactly one second of time headway. The lead vehicle is a puppet: at a scripted moment it will brake hard, at 6 m/s^2 , down to a standstill. It ignores our driver completely. Everything interesting in this chapter happens inside our driver's head.

Our driver does not know the script. What it has instead:

- **Senses.** It watches the lead vehicle the way a human eye does — as an optical silhouette that sits at some visual angle and grows or shrinks. It does not receive the gap in meters or the closing speed in m/s; it receives angles and their rates of change, with noise that shrinks as the target gets closer.
- **Beliefs.** A cloud of 75 simultaneous hypotheses about the current state of the world — where the lead is, how fast it moves, whether it is accelerating — each hypothesis weighted by how well it explains the last observation.
- **Preferences.** A quiet notion of how this drive is supposed to go: cruising near 10 m/s, pedals and steering mostly still, staying in lane, no collision ever, and a safety margin that would still work out even if the lead braked hard.
- **A plan.** A 6-second sequence of intended pedal and steering settings, currently: do nothing, keep cruising.

- **A surprise account.** A running total that fills with evidence that the current plan is no longer delivering the preferred future.

One rear-end event, from the authors' own simulation output (10 m/s, gap 10 m)



The event, moment by moment

t = 0.0 to 0.6 s — nothing happens, and that is almost the point. Steady following. The belief cloud tracks the lead within a whisker; the hypotheses about its acceleration hover around zero. The plan (keep cruising) delivers the preferred future in *most* rolled-out hypotheses — but not all. The imagined futures carry noise on the lead's acceleration, and over a 6 s horizon a fraction of them end too close or with a collision, so each step deposits a non-trivial amount into the surprise account even now: about 68 000 units per step in this condition, which is 7.6% of the re-plan threshold per step [OSF]. By the time the lead brakes, the account already stands at 0.31 of the way to a re-plan. The *realized* state — the gap and the speeds as they actually are — is comfortable; it is the *imagined* spread that is not silent. Keeping these two apart matters for everything that follows.

t = 0.8 s — the lead starts braking. The script fires: the lead's deceleration ramps toward -6 m/s^2 . In the very next belief update, the hypothesis cloud has already snapped to the new reality — the believed lead deceleration jumps to -2 m/s^2 with almost no spread. This is worth pausing on: the model has *detected* the braking essentially immediately. Detection is not the bottleneck.

t = 0.8 to 1.2 s — knowing is not yet acting. The driver knows the lead is braking, but its plan is still "keep cruising", and a plan is not abandoned just because the world changed — it is abandoned when it stops delivering the preferred future. Each new step now rolls the belief cloud forward and finds the planned future degrading: the predicted gap shrinks, the predicted safety margin erodes, predicted collisions appear among the hypotheses. The per-step deposit rises from 68 000 to 197 000 and then 267 000 [OSF]. The account climbs from 0.31 — 0.6 s of "the plan is going stale" that has nothing to do with sensory sluggishness, on top of the 0.8 s of pre-conflict drift that got it a third of the way there.

t = 1.4 s — the account is full: re-plan. The accumulated surprise crosses its threshold, and for exactly one timestep the model does the expensive thing: it discards the stale plan, generates a fresh set of candidate 6-second plans, scores each against the preferred future across the whole belief cloud, and keeps the best. The winner is unambiguous — brake, hard. The bottom panel of the figure shows this single re-plan event; there is precisely one in the whole trial.

t = 1.6 s — the brake reaches the wheels. The first step of the new plan executes: -6.2 m/s^2 . Response time, measured the way the paper measures it: 0.8 s from the lead's braking onset to our driver's brake onset — of which detection took at most one 0.2 s step, and essentially all the rest was evidence accumulation. Human rear-end response times in comparable staged conditions sit in the same range [Paper].

t = 1.6 to 3.4 s — riding it out. The lead stops; our driver brakes from 10 m/s to a standstill, easing off as the margin recovers, and comes to rest **2.05 m** behind the lead's bumper. No collision, no swerve — at this speed the preference landscape makes braking the cheap escape and steering the expensive one, which reverses at highway speeds (chapter 05). After the stop, the pedal trace wanders — the plan is incrementally patched rather than re-planned, and with both vehicles stationary the preferred future is satisfied almost no matter what the pedal does. The surprise account has gone quiet again.

What to take from this

- Detection and response are different things, and the model separates them.** The belief cloud caught the braking within one step; the response came 0.6 s later, when the *plan* — not the world — had accumulated enough evidence of failure. The way we read the human-factors literature, this matches it: drivers rarely miss that something moved; what takes time is concluding that their current course of action has stopped being adequate [Opinion].
- The account drifts even in steady following.** The accumulated quantity is an expectation over noisy imagined futures, and it is not zero inside comfortable following — it grades smoothly with the gap (chapter 05 gives the numbers across conditions). Response timing in a conflict therefore starts from a non-zero baseline that depends on the pre-conflict gap.
- One mechanism produced the whole episode.** Steady following, detection, response timing, braking style, and the come-to-rest margin all came out of the same loop — sense, believe, predict, evaluate, act, accumulate — with no per-phase sub-model. What that loop is made of, component by component, is chapter 03.

Where each part of the story is explained

Moment in the story	The machinery behind it	Chapter
Optical silhouette, noise shrinking with distance	looming perception	03, 08
The 75-hypothesis cloud snapping to the braking	particle-filter beliefs	06
"How this drive is supposed to go"	the preference prior	07
The account that fills with shortfall	residual of the pragmatic value, evidence accumulation	03, 05
One expensive re-plan	bounded, surprise-gated planning	03
Brake rather than swerve at 10 m/s	preference landscape vs speed	05

Notes for the mathematically curious

Level 1 — the accumulator. At each step the model computes how far the expected outcome of the *current* plan falls short of the best achievable expected outcome under the preference distribution — a nonnegative quantity, the residual information of the pragmatic value, written $\epsilon(t)$. It accumulates as $E(t) = E(t-1) + \lambda \cdot \epsilon(t)$; a full re-plan fires

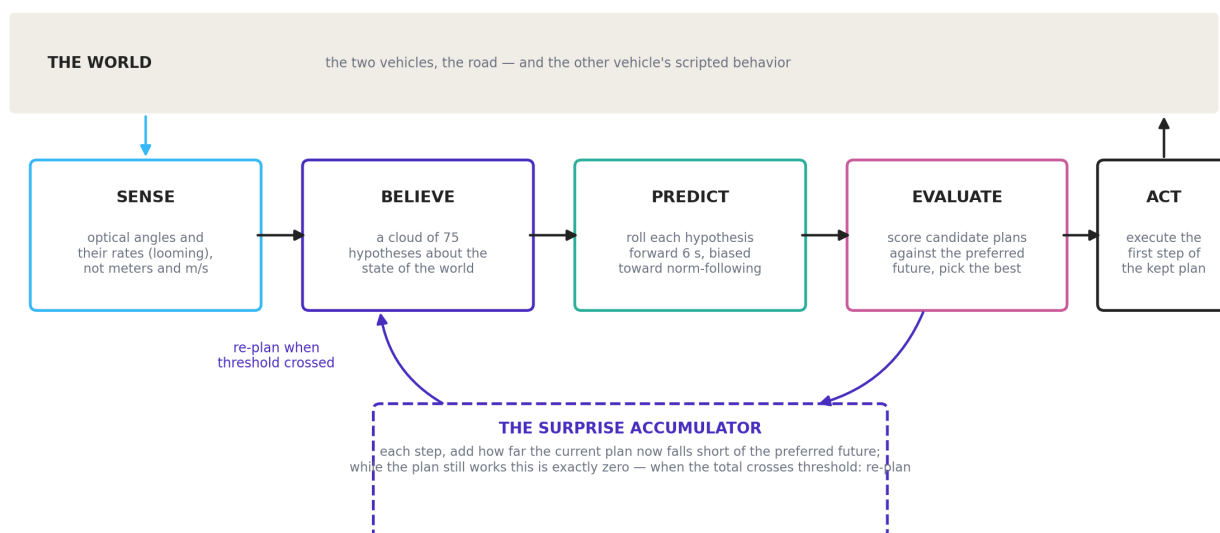
when $E \geq 1$, and E resets. λ (the paper's evidence-accumulation factor) is the single most response-time-sensitive parameter in the model [Paper].

Level 2 — this trial's numbers. Trial: Results_rear_end/Exp_7, seed 0, $\Delta t = 0.2$ s [OSF]. Lead braking onset $t = 0.8$ s (scripted countdown in the environment dynamics, ramp at 10 m/s^3 jerk to -6 m/s^2). Belief snap: the weighted mean of the 75 particles' lead-acceleration component moves $0.00 \rightarrow -2.00 \text{ m/s}^2$ between $t = 0.6$ and $t = 0.8$, with weighted standard deviation 0.00 — acceleration is directly observed with small noise in this configuration; the cloud's role is carrying *future* uncertainty, not present-state uncertainty (chapter 06). Re-plan flag: the executed policy differs from the extended reference policy only at $t = 1.4$ s (a_{cont} vs $a_{\text{cont_init}}$ in the deposit). First executed deceleration $\leq -1 \text{ m/s}^2$ at $t = 1.6$ s $\rightarrow$ response time 0.8 s. Final standstill gap 2.05 m (bumper to bumper). The evidence-accumulation equation is Eq. 13 of the paper; the preference terms it scores against are SI §2.4, Eqs. 44–52 [SI].

Chapter 3: what the model is

The loop

Everything the model does happens inside one loop, run five times per second (every 0.2 s). Chapter 02 showed the loop producing an event; this chapter names its parts.



Each pass: sense the world through looming vision, update a cloud of hypotheses about the current state, roll each hypothesis a few seconds into the future, check the current plan against the preferred future, execute the plan's next step — and deposit into the surprise account whatever shortfall the check revealed. When the account crosses its threshold, and only then, build a new plan from scratch.

The components, as input $\rightarrow$ output boxes

1. The world (the generative process). What actually happens: two vehicles with bicycle-model physics, a road, and a scripted other vehicle (chapter 06). The driver model never sees this directly. *Input:* both vehicles' controls. *Output:* the true state, advanced 0.2 s. [Code: dynamics_true.py per scenario]

2. The senses (looming perception). Converts the true state into what a human eye could report: the other vehicle's optical angle and its rate of change, plus own-vehicle signals (speed, pedal state, lateral position). Two consequences come free: measurement noise grows with distance, and expansion below a visual threshold (0.00215 per second [SI]) is effectively invisible — so detection distance emerges rather than being fitted. *Input:* true state. *Output:* a noisy observation vector. [Code: decoder.py, encoder.py]

3. Beliefs (the particle filter). The model's working memory: 75 parallel hypotheses ("particles"), each a complete candidate state of the world, each weighted by how well it explained recent observations. The cloud's *spread* is the model's honest uncertainty. When observations are sharp the cloud is tight (chapter 02's acceleration snap); when

the other vehicle's motives are ambiguous the cloud straddles the options — for example "will brake / will not" as two co-existing particle populations. Between observations the particles are *moved* by the same norm-shaped transition that prediction uses (component 4) — the swarm itself leans toward "others will behave", and observations correct it. *Input*: previous cloud + new observation. *Output*: updated cloud. [Code: `particle_filter.py`, `encoder.py`, `kde.py`]

4. Prediction (imagined futures). Every particle is rolled forward 6 s (30 steps) under the physics model. The other vehicle's imagined controls are not raw noise: at each step each particle holds a small tournament among candidate moves, weighted by norm compliance — futures in which the other vehicle behaves normally (stays in lane, keeps speed) win the sampling lottery, but only to the degree that its currently observed behavior has earned that trust. The norms are thus *inside the swarm's motion*, in the belief update and the roll-outs alike (chapter 06 makes this precise). *Input*: belief cloud + a candidate plan for our own controls. *Output*: a bundle of imagined 6-second futures. [Code: `dynamics.py`]

5. Preferences (the landscape). A description of how driving is supposed to go, encoded as a probability distribution over observations: comfortable speeds, gentle pedals, staying in lane, no collisions, and a counterfactual safety margin — "even if the lead braked hard right now and I reacted after one second, ordinary braking would still be enough". Six independent terms, each with interpretable knobs (chapter 07). *Input*: an imagined future. *Output*: a score of how preferred that future is. [Code: `reward.py` per scenario; SI §2.4]

6. The planner (bounded, not optimal). Holds one current plan — a 6-second control sequence. Ordinarily the plan is only *patched*: shifted one step, and only the newly exposed final step is re-optimized (ten rounds of one-step proposals). A full re-plan — propose 100 candidate plans, score them all against the preferred future across the belief cloud, refit the proposal distribution on the better half, repeat twenty rounds, keep the single best plan of the last round — runs only when the surprise account demands it. The budget is deliberately capped: the planner sometimes returns a merely decent plan, which is a modeling commitment about humans, not a bug [Paper]. *Input*: belief cloud, preferences. *Output*: the (kept or replaced) plan; its first step goes to the vehicle. [Code: `mpc_discrete.py`] *Revised 2026-09-03: the first edition said "the best tenth" and "~10 rounds". The released planner transforms its nominal elite fraction of 0.1 into 0.5 before using it, so 50 of 100 plans are elite, and it doubles its 10 iterations to 20 around every full re-plan; the Level 2 note gives the lines.*

7. The surprise account (evidence accumulation). The response-timing mechanism. Each step it receives the gap between what the current plan is now expected to deliver and the best that could be expected — zero only if every imagined future works out, which in practice it never quite does (chapter 02) — multiplied by a rate constant and added to a running total. Threshold crossed → full re-plan → total resets. *Input*: the plan's scored shortfall. *Output*: the re-plan trigger. [Paper Eq. 13]

Two currencies, one ledger

The scoring in component 5 has two parts, and their shared currency is the framework's main selling point:

- **Pragmatic value** — how well an imagined future matches the preferred one. This is goal-seeking: progress, comfort, safety margins.
- **Epistemic value** — how much an imagined future is expected to *teach* the model, by reducing the belief cloud's spread where it matters. This is information-seeking: looking, probing, easing off to see what the other driver does.

Both are measured in the same units and simply added, so caution and progress trade against each other without an arbitration rule. In the published collision-avoidance runs the pragmatic part dominates [Paper]; the epistemic part is the natural hook for glance behavior and uncertainty-driven slowing (chapter 08).

The looming channel, tested against human judgments

Added 2026-09-03. For our own purposes we asked which single scene quantity best predicts when people say they would intervene in a highway cut-in, shown as video clips frozen at successive moments (378 clips, 288 of them after the cut-in has begun, 10–26 raters per clip; one three-parameter probit threshold per candidate, scored out of sample by holding out each starting time-to-collision level in turn). The candidates were the gap, time to collision (equivalently inverse tau), the target's optical size, the required deceleration, and the optical expansion rate θ that the model's own perception stage computes. The expansion rate won outright: held-out weighted RMSE 0.113, against 0.152 for the gap, 0.152 for optical size, 0.168 for inverse tau and 0.289 for required deceleration, with the sampling-noise floor of the data at 0.118. A two-dimensional rule fitted freely in log gap and log time-to-collision put a weight of 0.497 on the gap (0.474–0.512 across folds), which is the small-angle identity $\theta \approx W \cdot \Delta v / \text{gap}^2$ rather than an approximation of it

[Study].

We report this because it was not the question we set out to answer, and because it cuts the other way from what chapter 07 reports about the preference terms: the model's *perceptual* front end, chosen by the authors on visual-control grounds, is the quantity that human intervention judgments track best on this paradigm. One caveat belongs beside it. Xue, Markkula, Yan and Merat (2018, *Accident Analysis and Prevention* 118, 114–124) compared the same two looming cues on brake onsets in a driving simulator and found inverse tau the better cue for both threshold and accumulator models, with accumulator models fitting the response-time distributions better than pure thresholds. Whether the reversal is the paradigm (frozen video without self-motion against a simulator), the task (a judged intervention point against an executed brake onset) or the scenario (a cut-in against a braking lead) is open; naturalistic onsets would settle it [Opinion].

New in this version, 28 September 2026: not in the edition you received. Naturalistic onsets now point to the task or the paradigm. On 2,208 real closing cut-ins in the highD dataset, whether the follower brakes within 3 s is predicted slightly better by inverse time to collision than by the expansion rate (area under the ROC curve 0.739 against 0.713; difference 0.026, 95% interval 0.006 to 0.043), at every braking threshold we tried and at short and long gaps alike. So video judgments follow the expansion rate, while executed braking in real traffic follows inverse tau, as Xue et al. found in the simulator [Study]. The model's perceptual front end and its inverse-tau preference may each be right for its own job [Opinion].

What is deliberately human about it

Design choice	The human claim behind it
Looming instead of range sensors	drivers see angles, not odometry; distant threats are genuinely harder to perceive
A particle cloud instead of one estimate	drivers entertain multiple readings of an ambiguous scene at once
Norm-shaped prediction	drivers expect others to behave; trust is withdrawn when observed behavior stops earning it
A capped planning budget	drivers satisfice; an optimal planner reproduces the wrong behavior
Surprise-gated re-planning	drivers do not continuously re-decide; they act when evidence has built up that they must

What it is not

- It is **not learned from data**: no training set, no fitted network. Thirteen parameters were hand-tuned [Paper]; the rest is structure.
- It is **not an optimal controller**: bounded planning, noisy perception, and normative trust are all deliberate departures.
- The other vehicle is **not intelligent**: it follows a script and never reacts to our driver (chapter 06). There is no negotiation or interaction in the published model.
- It is **not fast**: the code is written for GPU; on a CPU, one simulated timestep of a batched run has cost us anywhere from under a second to tens of seconds, and the variation between runs is wide with no clean predictor (a 45-step scenario with four parallel repeats has taken between minutes and hours on the same machine). Plan batches to be restartable, and measure before extrapolating [Study].

Notes for the mathematically curious

Level 1 — the loop as equations in words. Perception: particle weights are multiplied by the likelihood of the new observation under each particle, then normalized; particles are refreshed by resampling when weights degenerate. Prediction: particles are pushed through the bicycle dynamics with control noise on the other agent, reweighted by norm-compliance (chapter 06, Level 2). Planning: cross-entropy method over control sequences — sample, score by expected free energy, keep the elite half, refit the sampling distribution, iterate, and return the best plan of the final round. Acting: execute the first control of the incumbent plan.

Level 2 — sizes and references. $\Delta t = 0.2$ s; horizon $H = 30$ (6 s); particles $N = 75$; CEM: 100 candidate plans; elite set 50 — the planner's constructor raises the nominal $\text{top_percent} = 0.1$ to $0.1^{(\log 0.5 / \log 0.1)} = 0.5$ and takes $\max(\min(50, N/2), \lceil 0.5 \cdot N \rceil)$ plans, so half of them at $N = 100$; iterations 10 for the one-step patch and 20 for a full re-plan (iters is doubled around each `generate_optimal_plan` call, so the SI's 20 and the paper's 10 are both right,

for different calls) [Code: `mpc_discrete.py`; revised 2026-09-03]; looming threshold 0.00215 s^{-1} , a fixed value — the decoder also carries an unused distance-dependent threshold, interpolating 0.00377 s^{-1} at 20 m to 0.00215 s^{-1} at 40 m headway, which the belief update computes and then overrides with the fixed one [Code: `decoder.py`, `transform_to_looming`; `encoder.py`]; evidence accumulation $E(t) = E(t-1) + \lambda \cdot \varepsilon(t)$, re-plan at $E \geq 1$ (Paper Eq. 13). Observation model and looming transform: SI §2.2; belief update §2.3; preference terms §2.4 Eqs. 44–52; planner §2.5 [SI]. Parameter values as shipped: the OSF deposit's `Setups_*.xlsx` (65 columns per run) [OSF].

Chapter 4: the scenario playbook — what actually changes, and the switching checklist

This chapter's core evidence is a column-by-column diff of the authors' own setup tables for all three scenarios ('Setups_.xlsx' in the OSF deposit, 65 parameters per run) [OSF], plus a file-level diff of the per-scenario code [Code].**

The headline: the driver barely changes; the world does

The published model runs three scenarios: **rear-end** (a braking lead vehicle), **oncoming** (an opposite-direction vehicle drifting into our lane), and **intersection** (a crossing vehicle turning across our path — the held-out one). Diffing the authors' own setup tables across all baseline runs of the three scenarios gives an unambiguous answer to "what is different, exactly": of 65 configuration parameters, everything describing the *driver* — perception noise, looming threshold, preference weights, planning budget, evidence accumulation, particle counts — is **identical across all three scenarios**, with exactly one exception described below. What changes is the *world*: geometry, initial speeds, and the other vehicle's scripted behavior [OSF].

The same conclusion appears at the file level. Each scenario is a package of three files, and only three [Code]:

File	Job	In plain terms
<code>dynamics_true.py</code>	the world's actual physics and the other vehicle's script	what really happens
<code>decoder_true.py</code>	how the true state becomes the driver's observations	what can be seen
<code>reward.py</code>	the preference terms and norms that are scenario-shaped	what counts as normal here

Everything else — the particle filter, the planner, the looming transform, the evidence accumulator — lives in `src/common/` and is shared untouched. The way we read it, this is the model's strongest structural claim: *one driver, many worlds* [Opinion]. It is also what makes the held-out intersection test meaningful — the driver that handled the rear-end scenario was dropped into a new world, not re-engineered for it.

The full diff, in five groups

1. Stage-setting (different by definition). Initial positions, speeds, headings, the desired speed, and the episode length. Rear-end: both at 10–25 m/s in the same lane, gaps giving 0.67–3.5 s headway. Oncoming: both at 17.88 m/s (40 mph), the other vehicle 202.7 m away in the opposite lane. Intersection: ego approaching at 14.7–16.3 m/s from 36–40 m out; the crossing vehicle either waiting at the junction or rolling at 6.8 m/s [OSF].

2. The other vehicle's script (the heart of a scenario). Rear-end: a countdown, then brake at a scripted intensity (the per-condition `a_tar_min_intensity` and `brake_intensity` columns). Oncoming: a time-to-collision trigger (`ttc_trigger`), then a lateral incursion to a target intrusion depth (`rel_target`) — implemented, notably, as the script *optimizing* its own steering trajectory to hit that target. Intersection: a TTC-triggered turn across our path. These script parameters exist **only** in their own scenario's table — they are the scenario [OSF] [Code].

3. The one driver-side change: assumed steering variability. `w_sd_model` — how much steering wobble the driver's internal model attributes to the *other* vehicle when imagining its futures — is 0.0045 in rear-end and 0.4575 in both lateral scenarios, a factor of one hundred [OSF]. This is not a perception setting; it is an assumption inside the driver's head about what kind of agent it is facing: a lead in a queue does not steer, an oncoming or crossing vehicle might. It is the single number by which the driver was told what type of situation it is in. Whether a future version could *infer* this rather than be told is, to us, an open and interesting question [Opinion].

4. Scenario-shaped preference and norms (inside `reward.py`). The preference terms for speed, pedals, and collision are shared; what is scenario-shaped is the **lane structure**: what "in my lane", "in the oncoming lane", and

"off the road" mean geometrically, and what the *other* vehicle counts as doing normally (its norms — chapter 07 details all three). Rear-end's version even contains hand-built bookkeeping that penalizes dawdling mid-lane or aborting a lane change; oncoming's version treats hard braking by the other vehicle as a norm violation alongside leaving its lane; intersection's draws the geometry of running a red light and cutting the corner [Code].

5. Calibrated safety assumptions. The safety-margin preference asks "would ordinary braking still save me if the other vehicle did its worst?" — which requires assuming what "its worst" is. That assumed worst-case deceleration is calibrated per scenario against a free-following study rather than shared [SI]. One property of the shipped calibration worth knowing: the lookup table covers steady-state headways up to about 1.0–2.1 s depending on speed, and outside that range the interpolation clamps to the table edge, so for the longer-gap rear-end conditions the assumed worst case saturates at -8 m/s^2 [Code: `Analysis_following.xlsx`, `find_parameters`]. Chapter 10 returns to this as a general lesson about calibration coverage.

The three files, line by line

Added 2026-09-03, from a line-level diff of `src/rear_end_test/`, `src/oncoming/` and `src/intersection/` [Code]. The five groups above say what changes; this section says exactly where, because a new scenario will be written by editing these three files and it matters which lines are the scenario and which are the driver.

A naming trap first. The intersection scenario's files are called *side* at top level (`simulation_side.py` sets `name = 'intersection'`, and `Analysis_side.py` analyzes it) but *intersection* inside `src/`. Searching for one name finds half the code.

`decoder_true.py` **is identical in all three.** The three versions differ by exactly three lines, all of them docstring, where the rear-end version documents two extra state variables (`t_brake_tar`, `j_brake_tar`) belonging to its target's brake script. How the world becomes observations — the looming transform, the noise model, the gaze gate — is scenario-independent in the released code. The only thing that makes it scenario-dependent is the vehicle dimensions it is constructed with, since the looming angle depends on the target's width.

`dynamics_true.py` **is where nearly all the code volume is, and none of the driver.** The three versions are 162, 354 and 665 lines (rear-end, intersection, oncoming), and the spread is entirely the other vehicle's script:

scenario	what <code>forward_state_tar</code> does	supporting machinery
rear-end	counts down to <code>t_brake_tar</code> , then ramps deceleration at <code>j_brake_tar</code>	none; a few lines of arithmetic
intersection	steers the target through the turn geometry	<code>tar_loss</code> , <code>get_tar_steering_rate</code> , <code>load_steering_rate</code> : solves for the steering rate that tracks the intended path
oncoming	drives the target along a synthesized incursion trajectory	<code>propagate_forward</code> , <code>cost_function</code> , <code>ctrl_gradient_descent</code> , <code>optimize_control_part</code> : a gradient-descent optimal-control solve

The trap in this file is the word "cost". `src/oncoming/dynamics_true.py` contains a `cost_function`, and it has nothing to do with the driver's preferences: it is the scenario author's own objective, penalizing deviation of the target's lateral position from a reference path plus a terminal mismatch, minimized by gradient descent to manufacture a smooth incursion that reaches the scripted depth at the scripted moment. Reading it as part of the driver model would be a serious misunderstanding; it is stage machinery, not psychology.

`reward.py` **is the only file where the scenario changes the driver**, and the three versions differ in exactly three places:

- The lane-geometry mapping, in `compute_features`.* A handful of lines that turn a raw lateral coordinate into "how far am I from where I should be". Rear-end and intersection share one version, which treats the ego as being on a two-lane road with a viable lane to the left and snaps near-zero offsets to exactly zero. Oncoming needs a different one because the lane to the left is the *oncoming* lane rather than a free overtaking lane, so it distinguishes `in_other_lane` from `left_lane` and maps them separately. This is checklist item 4, and it is roughly six lines.
- A lane-change bookkeeping block that exists only in rear-end.* About twenty lines that track how long the ego has been between lanes, add a heavy penalty for an *aborted* lane change (returning to the lane it started in after more than 1.5 timesteps of straddling) and a growing penalty for dwelling between lanes beyond nine timesteps. Oncoming and intersection delete it. As we read it, this is hand-built shaping rather than principle, a patch to stop the planner from discovering that hovering on the lane line is comfortable, and anyone porting the model should know it is there and decide deliberately whether to keep it [Opinion].

- 3 `get_weights`, *the norms, which is the real scenario content*. This function is the driver's prior over what the *other* vehicle will normally do; the dynamics call it as `normative_probability` when propagating beliefs about the target (chapter 06). All three versions differ completely, and the progression is instructive: rear-end scores position only (in its lane, weight 1; just outside, `weigh_particles`; off road, `weigh_particles × full_violation_factor`); oncoming scores position relative to the oncoming lane center **and speed**, through the quadratic speed norm $1 - 2.25 \cdot (v/v_0 - 1)^2$ folded in by a minimum, with the code's own comment that braking is to be treated as norm-violating like leaving the lane; intersection uses a two-dimensional road-geometry mask with a quarter-circle corner of radius $10 + \frac{1}{2} \cdot \text{lane width} + \frac{1}{2} \cdot \text{vehicle length}$, plus a term that down-weights the target running the red light. Stay in your lane; then also keep your speed; then also stay on the road and obey the light. This is where the modeling judgment lives, and it is written by hand each time.

Two smaller differences: rear-end keeps several selectable looming-reward variants (v2–v7, and a non-looming fallback) where the two lateral scenarios hard-code v7; and oncoming guards its braking-feasibility term with a check that the two vehicles are traveling in the same direction, which rear-end does not need. Parameters are almost entirely shared, with `lane_width` = 3.65 m in rear-end and oncoming against 3.5 m in intersection, and the one driver-side change already noted above (`w_sd_model`).

The switching checklist

To move the model to a new scenario — a cut-in, a cyclist overtake, a pedestrian crossing — these are the decisions, in the order we would take them. Items 1–3 are mechanical; items 4–7 are modeling judgments that deserve explicit argument in any write-up.

- 1 **Stage the world.** Geometry, lanes, initial states, desired speed, episode length. (Setups columns; scenario `dynamics_true.py`)
- 2 **Script the other agent.** Trigger condition and maneuver, with its intensity as a sweepable condition parameter. (`dynamics_true.py`)
- 3 **Check observability.** Does the driver see the new agent through the same looming channel? A pedestrian subtends different angles than a truck; the decoder's geometry (vehicle dimensions) must match. (`decoder_true.py`) Note also that the released applicability tests are binary: looming is perceived only while the target is ahead and within 3 vehicle widths laterally, and the collision, safety-margin and closing-rate terms apply only within 1.15 widths, so a target that is *partly* in the lane is all-or-nothing to the driver [Code: `decoder.py::test_looming_viability`]. None of the three released scenarios sustains that state; a cut-in does, for about 2.5 s in recorded clips [Study] (chapter 07 says what follows).
- 4 **Draw the lane structure into the preferences.** Define what lateral positions mean — own lane, oncoming, shoulder — for *this* road. This is hand geometry today; there is no map format. (`reward.py`, lateral term)
- 5 **Write the other agent's norms.** What does *normal* behavior look like for that agent here, as regions in its state space with graded compliance weights? This is the most judgment-heavy step, and it directly shapes how paranoid or trusting predictions are. (`reward.py::get_weights`; chapter 06)
- 6 **Set the assumed variability.** Choose `w_sd_model` (and its longitudinal sibling) for the new agent type — the "what might this thing do" dial. (item 3 above)
- 7 **Re-calibrate the safety assumption.** Fit the assumed worst-case deceleration for the new context, and confirm the calibration covers the intended speed and headway range (item 5 above). ([SI])
- 8 **Define the measurements.** Response-onset definition, collision definition, and the condition grid, so results are comparable with the published ones. (analysis scripts)

The honest summary of effort: steps 1–3 are days, steps 4–7 are where the scenario's scientific content lives, and skipping the argument for any of 4–7 produces a model that runs but persuades nobody [Opinion].

Notes for the mathematically curious

Level 1 — why one driver can serve three worlds. The agent's objective (expected free energy over a 6 s horizon) never mentions the scenario; scenario enters only through (i) the generative process being simulated, (ii) the lateral/lane geometry inside the preference distribution, and (iii) the norm-compliance weighting inside the prediction step. Since all three are data to the same algorithm, transferring the driver is exactly transferring those three objects.

Level 2 — the diff, precisely. Baseline-run comparison across the three Setups tables: differing columns are {initial state: `x_ego`, `v_ego`, `x_tar`, `y_tar`, `theta_tar`, `v_tar`, `v_ego_des`, `T`}, {script: `a_tar_min_intensity`,

brake_intensity (rear-end); ttc_trigger, rel_target (oncoming); ttc_trigger (intersection)), {driver-side: w_sd_model = 0.0045 vs 0.4575}, and num_plans = 1000 in oncoming's extended-budget sensitivity runs vs 100 baseline. All remaining ~50 columns are identical across scenarios, including every preference weight (v_ego_sd_des, a_ego_sd_des, w_ego_sd_des, lane_change_cost, road_leave_cost, collision_cost), every perception noise, the looming threshold, N_norm, H_norm, alpha, and the evidence-accumulation settings (EA_mode, EA_fac = λ , EA_init) [OSF]. Oncoming's target script solves a small optimal-control problem for the incursion (gradient descent on a steering sequence to reach the scripted lateral target at the scripted time; src/oncoming/dynamics_true.py) [Code].

Chapter 5: normal driving versus critical events

There is no emergency mode

The most useful single fact in this chapter: the model has **no mode switch**. No flag flips from "normal" to "critical"; no emergency sub-model wakes up. The loop of chapter 03 runs identically at every timestep of every drive. What differs between everyday following and a hard conflict is *which parts of the machinery are doing the work* — the same objective, evaluated in a different region of the landscape. That is a substantive claim about drivers, inherited from the zero-risk tradition: emergency response is ordinary regulation, pushed hard.

The regimes still look completely different from outside, and it is worth being precise about why.

Normal driving: the quiet regime

In steady following (chapter 02, $t < 0.8$ s), four things characterize the model's state:

- **The surprise account drifts, slowly.** The current plan delivers the preferred future in most imagined rollouts, but the imagined spread always contains a few futures that end too close, so something accumulates even in steady following: in the authors' own runs, from 2% of the threshold per 0.8 s at a 3.5 s gap to 44% at a 0.5 s gap, all of it from the collision and safety-margin terms [OSF]. Extrapolated, the accumulator would re-plan on its own after 2–7 s of uneventful following at gaps of 2 s or less; the published simulations do not show this because they start 0.8 s before the lead brakes. How much the pre-conflict drift matters for response timing turns out to be small when a run opens a few seconds before a conflict at ordinary headways: across 23 rear-end scenarios replayed from recorded lead-vehicle profiles (1.3–35.5 m/s), an open-loop surrogate of the accumulator that starts its account at zero reproduced the closed loop's brake onsets to a median 0.30 s, while giving the surrogate a half-full account at the start, as if the driver arrived mid-cycle, over-corrected by about a second. The drift is a real property with small near-conflict consequences; designs with long benign run-ins are where it would bite [Study].

New in this version, 28 September 2026: not in the edition you received. In sustained following the drift does trigger re-plans, and the re-plans brake. Behind a lead holding a constant speed, the released configuration began braking after 3.2 s at a 1.5 s headway and after 4.6 s at 2.0 s, with or without gaze choice and with the scenario's own scripted lead; some repeats braked to a standstill; with perception noise raised a hundredfold it followed steadily. Behind real leads replayed from the highD dataset, in 18 five-second episodes in which the lead's acceleration never exceeded 0.5 m/s^2 in magnitude, it braked by at least 1 m/s^2 in 56% of runs, typically 2.5–5 s in and often at -6 to -7 m/s^2 , while the drivers of the same episodes did not brake at all (their lowest acceleration -0.63 m/s^2). The episodes were staged with the released calibration (find_parameters), which returned the same values for all 18 (desired-speed offset 0, assumed lead braking -8 m/s^2) [Study]. The published scenarios never exercise this regime, and nothing in them is wrong because of it; it matters for any use of the model with a run-in longer than a few seconds. What makes braking win at the first re-plan, and why noisier perception suppresses it, we have not established [Opinion].

- **Planning is incremental.** The plan is shifted and cheaply patched each step; the expensive candidate-generation machinery is dormant. Most timesteps of a normal drive never trigger a single full re-plan.
- **Trust is extended.** The other vehicle has been behaving normally, so predictions concentrate on norm-following futures; the long tail of "what if they do something wild" is present but carries little weight (chapter 06).
- **Behavior is shaped by the gentle terms — in the realized state.** Speed preference, pedal smoothness, and lane centering — the low-stakes preference terms — are what the driver's actual state is scored against, and on that state the collision and safety-margin terms are satisfied. One consequence of the published design is worth noting: with the lead braking 0.6 s into each run and the driver following its fixed reference plan until the first

re-plan, what "normal driving" looks like as *behavior* in this model is exercised only briefly in the published simulations [Code: EA_init = False].

What normal driving *is*, in this model: the region of the state space where preferred futures remain reachable without doing anything unusual. Keeping the vehicle inside that region is what the quiet machinery continuously accomplishes.

A critical event: the same machinery, loud

When the world breaks the pattern (chapter 02, $t \geq 0.8$ s), each of the four quiet properties inverts, in a fixed causal order:

- 1 **Trust is withdrawn first.** The other vehicle's observed behavior stops matching its norms; the norm-conditioning releases the prediction tail, and imagined futures fan out to include the bad ones. This happens in the belief/prediction machinery *before* any decision is made — expectation revision precedes action, as the predictive-processing account says it should.
- 2 **The heavy preference terms wake.** Predicted futures now touch collision and eroded safety margins; the terms that were quiet begin to dominate the scoring.
- 3 **The surprise account fills.** The incumbent plan's expected outcome falls short of the best achievable; the shortfall is deposited step by step. This is where response *time* comes from — not from sluggish senses but from evidence integration.
- 4 **One expensive re-plan fires,** and the chosen escape is executed with the same bounded planner as always.

The choice of escape is a property of the landscape, not a rule

Nothing in the code says "brake below 60 km/h, steer above". Yet the authors' own runs show exactly that structure emerging [OSF] — across the baseline rear-end grid:

Initial speed	braking only	steering involved	still behind the lead at rest
10 m/s	86%	0%	the norm
15 m/s	34%	54%	mixed
20 m/s	1%	96%	rare
25 m/s	0%	40% (most leave the road instead)	rare

The gradient has a physical reading: at low speed, comfortable braking removes the kinetic energy in time and never violates the pedal-smoothness preference much; at high speed the deceleration needed to stop behind the lead becomes so severe that a lane change, despite its own preference costs, scores better. The maneuver choice falls out of comparing imagined futures under one preference landscape.

The 25 m/s row deserves a note: at the highest speed, a majority of the deposited runs (58% averaged over the gap grid; 72% at the shortest gap and 53% at the longest) end by leaving the road, mostly during the avoidance maneuver and mostly to the left, through and beyond the adjacent lane [OSF]. At a 3.5 s gap moderate braking would suffice.

New in this version, 28 September 2026: not in the edition you received. (*Reworded; the edition you received read: "we read these departures as a property of the lane-change control at speed rather than as a deliberate trade-off [Opinion]"*.) Whether these departures are best read as a property of the lane-change control at speed or as a trade-off the preference landscape makes on purpose is a question of reading on which we would defer to the authors [Opinion]; either way, analyses at 25 m/s should track road departure as its own outcome class.

What this means for using the model

- **Response time is not a parameter you set** — it is a prediction that emerges from the interplay of perception noise, norm trust, the accumulator rate, and the preference landscape. To move it, you move those (chapter 09 lists which moves what).
- **New in this version, 28 September 2026: not in the edition you received. The response-time relation holds in direction on real motorway data.** On 1,033 lead-deceleration events in the highD dataset, the log of the follower's response time rises with the log of the initial time gap, and more steeply the harder the lead brakes: slopes +0.23, +0.40 and +0.45 for mild, moderate and hard lead decelerations, against +0.59 in the deposited runs. Real responses are about three times slower (median 2.2–3.1 s at 1–2 s gaps against 0.8 s). Most of that is

the data: few motorway events are emergencies (32 leads braked at 3 m/s^2 or more), and real lead braking takes about 2 s to build up where the scenario's takes a tenth of that. With so few hard events this is descriptive, not a test [Study].

- **The normal-driving regime is not free** — the same machinery that produces crisp emergency behavior must also idle plausibly, and misconfigured preferences show up first as fidgety normal driving.

Notes for the mathematically curious

Level 1 — one objective, two regimes. The expected-free-energy score of the incumbent policy decomposes over preference terms. In the quiet regime the per-step residual ϵ — best-achievable minus incumbent *expected* pragmatic value — is small relative to the loud regime but not zero: the expectation runs over 75 noisy rollouts $\times$ 30 steps, and the collision and safety-margin terms of the worst few dominate it (5×10^3 to 10^5 per step in the deposit). E therefore drifts at rate $\lambda\epsilon$ even before any event. In the loud regime ϵ jumps by a factor of 3–4, E integrates it, and the re-plan at $E \geq 1$ is a drift-diffusion first-passage with a model-supplied drift and a non-zero starting point. Maneuver choice is the argmax over candidate policies of the same score; no separate decision rule exists.

Level 2 — the numbers above. Maneuver mix: mean over the 28 baseline rear-end conditions grouped by initial speed, from the authors' `Analysis_rear_end.xlsx` (braking_post, overtaking_post, brake_steer_post, leave_road, collision) [OSF]. Accumulator drift in benign following: per-step ϵ from the deposited pragmatic-value components (v arrays), aggregated per condition over the pre-onset window [OSF].

Chapter 6: other agents, and what the driver believes about them

Two completely different questions, easily conflated

"What is the other vehicle doing?" has two answers in this model, produced by two unrelated pieces of machinery, and keeping them apart is the key to reading the code and the paper correctly:

- 1 **What the other vehicle actually does** is decided by a *script* — simple, deterministic, and entirely outside the driver model.
- 2 **What the driver believes the other vehicle is doing and might do next** is decided by the belief machinery — a particle swarm whose own motion is norm-shaped, serving both the tracking of the present and the imagining of futures.

The particle filter is *not* how the other vehicle's behavior is generated; it is how the driver's uncertainty about that behavior is represented. The script answers question 1; the particle filter and the prediction roll-outs answer question 2.

Side one: the script (the truth)

Each scenario's `dynamics_true.py` [Code] contains the other vehicle's entire behavioral repertoire, and it is deliberately primitive:

- **Rear-end:** drive at constant speed; when a countdown expires, ramp braking to a scripted intensity; stop. The intensity and timing are condition parameters swept across runs [OSF].
- **Oncoming:** drive straight in the opposite lane; when time-to-collision falls below a trigger, steer into our lane to a scripted intrusion depth. The incursion trajectory is computed by a small optimizer so it hits the scripted depth at the scripted time — a fancier script, but still a script [Code].
- **Intersection:** approach or wait at the junction; on a trigger, turn across our path.

Three properties matter more than the details. The script **never reacts to our driver** — no negotiation, no yielding, no eye contact; every published result is about unilateral avoidance, and any interaction study would need this replaced [Paper]. The script is **where a scenario's difficulty lives** — sweeping its intensity and timing generates the condition grids. The script is **invisible to the driver**, who meets it only through optics.

Side two: the beliefs (the driver's honest ledger)

The present: a cloud of 75 hypotheses. The particle filter maintains 75 complete candidate states of the world — each one a full "it could be like this": positions, speeds, headings, and, crucially, the other vehicle's current acceleration and steering. Each observation re-weights the hypotheses by how well they explain it; hypotheses that keep failing get culled and replaced near ones that succeed. The cloud's spread *is* the model's uncertainty — no separate confidence number exists.

Between observations, each particle is *moved* by the norm-shaped transition described in the next section — the swarm drifts toward "that vehicle will keep behaving", and each arriving observation then corrects the drift toward what actually happened. Chapter 02 showed the resulting subtlety: in the rear-end configuration the observation channel is sharp enough that the cloud tracks the present tightly (the believed lead deceleration snapped to the truth in one step). The cloud earns its keep elsewhere — under ambiguity. When an oncoming vehicle wanders near the lane line, hypotheses compatible with "drifting but staying" and "beginning an incursion" coexist with real weight on both, and the driver's planning sees *both* futures. A single-best-estimate tracker is structurally unable to be of two minds; the particle cloud is of two minds routinely, which we read as one of the model's most human features [Opinion].

Where the norms live: inside the particle swarm's own motion

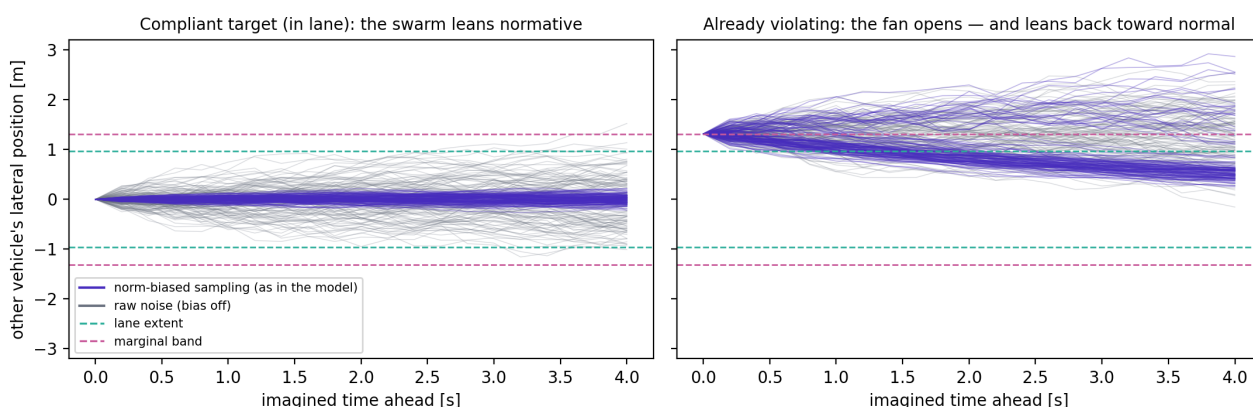
This section exists because it answers a question we ourselves initially got only half right: the normative machinery is not a layer applied *on top of* predictions — **it is built into how every particle moves**. We have checked the following against the code line by line [Code], and it holds for the belief update and the planner alike, because both are served by literally the same transition object (`src/common/dynamics.py::forward_tar_agent`; attached to the norm weights once, in the simulation setup).

Whenever any particle's picture of the other vehicle has to advance by one timestep — which happens both when the belief cloud is updated between observations and when futures are imagined during planning — the particle does not simply add random noise to the other vehicle's controls. It runs a **mini-tournament**:

- 1 Propose 32 candidate next moves for the other vehicle (its current controls plus random variation, sized by the `a_sd_model / w_sd_model` dials).
- 2 Score each candidate by norm compliance in three snapshots: where the vehicle *is* now, where the candidate puts it *one step* ahead, and where the candidate would put it *four seconds* ahead if held. The "soon" and "later" scores are averaged; the overall score is the *worse* of "now" and "that average" — a candidate that looks fine now but drifts off the road later is tainted by its future.
- 3 Draw **one** winner by lottery, with tickets proportional to the scores. That winner becomes the particle's next state.

So a compliant lead vehicle's particles overwhelmingly "choose" futures that stay in lane at speed — the swarm itself leans normative. Norms are the sampling bias of the swarm's own dynamics, not a filter applied afterward.

The norm tournament inside each particle's motion (illustrative lateral-only reimplement of the sampling rule)



The trust cap falls out of the same arithmetic, elegantly. The "now" score in step 2 is the same for all 32 candidates — it describes where the vehicle already is, which no candidate can change. While the vehicle is behaving normally that shared score is high, so the *differences* between candidates (their futures) decide the lottery, and the normative bias bites (left panel of the figure: the biased fan hugs the lane while raw noise sprays). The moment the vehicle is observed grossly misbehaving, the shared "now" score collapses, becomes the ceiling for

every candidate at once, and the lottery goes momentarily near-uniform: the bias dissolves and the swarm fans out over everything the vehicle could physically do. Trust is not a separate mechanism with its own parameter; it is the `min()` in step 2 doing its work. Revoked on evidence, within a step or two.

The right panel shows a second-order consequence we verified in the arithmetic and think is worth knowing: for a violating target the fan opens, *and* it leans back toward the lane — any hypothesis that wanders back into the normal region regains its bias and is recaptured. The swarm simultaneously entertains the long tail and expects the violator to eventually return to normality, which strikes us as a rather human expectation to hold [Code, our reading].

The same tournament, two different jobs. In the *belief update* the tournament runs for one step and is immediately corrected by the next observation — so norms gently shape where the cloud drifts between glances at the world, and reality has the last word. In *planning roll-outs* the tournament runs 30 steps with no observations to correct it — so norms shape the entire 6-second fan of imagined futures, which is where they influence decisions. One mechanism, two exposures; the second is where the behavioral consequences (relaxed following, late-but-not-too-late alarm) come from.

Added 2026-09-03. A third exposure appears when the observation channel is closed: **coasting through an occlusion**. When we forced off-road glances through the code's own gaze gate (chapter 08), the same norm-shaped transition carried the cloud forward essentially uncorrected, and everything downstream kept consuming it: a driver who saw the lead start to brake and then looked away kept *inferring* the conflict's development from the coasting belief, kept accumulating evidence, and could commit to braking mid-glance. The belief machinery is not a passive sensor buffer; it is a short-horizon simulator that runs with or without fresh input [Study].

Two dials size the raw variation the tournament chooses among (`a_sd_model`, `w_sd_model`): how much acceleration and steering wobble the driver attributes to "vehicles in general". Chapter 04 showed the steering dial is the one number that distinguishes the driver across scenarios (0.0045 for a queueing lead, 0.4575 for oncoming and crossing traffic) [OSF]. Without the tournament, raw noise alone would make the driver either paranoid (every future includes wild swerves) or oblivious (noise too small to cover real incursions); the norm bias with its built-in trust cap is the model's resolution of that dilemma [Paper].

Why this arrangement is worth copying

- **Detection speed comes from expectation violation, not from tuned vigilance.** The driver is relaxed precisely because compliant futures are weighted up; it becomes alert precisely when compliance visibly fails. One mechanism covers both, with no free "alertness" parameter.
- **The uncertainty is decision-grade.** The same cloud that represents "they might brake / might not" is what plans are scored against, so caution scales automatically with genuine ambiguity — no hand-tuned safety buffer.
- **The seams are explicit.** Everything the driver assumes about others is localized in three places: two noise dials, one norm geometry, one trust cap. Chapter 09 uses exactly these seams for modifications (an inattentive-driver prior, a trusting-driver variant, a pedestrian norm set).

The limits, stated plainly

The other agent has no mind: it does not perceive our driver, and the driver's model of it contains no recursive "they see me seeing them". Cooperative and communicative phenomena — gap negotiation, hesitation reading, signaling — are outside the published model's scope [Paper]. The norms are hand-drawn per scenario, which is honest but does not scale; learning them from data seems to us one of the most natural extension projects this line offers [Opinion].

Notes for the mathematically curious

Level 1 — the filter. Standard sequential importance resampling: predict each particle through the dynamics, weight by observation likelihood, normalize, resample on weight degeneracy. Later papers in the line replace raw particles with a kernel-density / Gaussian-mixture representation so beliefs can move outside the initial particle support. Prediction for planning: for each particle, sample other-agent control noise, roll the joint state 6 s; the resulting bundle approximates the predictive distribution over futures conditional on our candidate plan.

Level 2 — the tournament as implemented (`src/common/dynamics.py::forward_tar_agent`) [Code]. Per particle and per transition: draw $N_{\text{norm}} = 32$ control perturbations $\sim N(0, a_sd_model/w_sd_model)$, scaled by

noise_pred_fac = 0.2 when called in prediction mode; for each candidate compute compliance weights from the scenario's norm geometry (`reward.py::get_weights`) at the current state (`w_now` — note: identical across candidates), one step ahead (`w_next`), and `H_norm = 20` steps ≈ 4 s ahead under held controls (`w_long`); combine `w_future = harmonic mean(w_next, w_long)`; overall `w = min(w_now, w_future)`; normalize over the 32 candidates and sample one index from the categorical — the sampled candidate becomes the particle's next target state. Compliance weights: 1 inside the normal region, `weigh_particles = 0.001` marginally outside, times `full_violation_factor = 0.01` for gross violations [OSF]. The trust cap is the `min()` with the candidate-independent `w_now`: once `w_now < w_future` for all candidates, every candidate carries the same weight and the categorical is uniform — the bias vanishes without any dedicated switch. The same `Dynamics` instance is passed to both the Encoder (belief update; one call per step, observation-corrected via the KDE update) and `BeliefDynamics` (planner roll-outs; 30 uncorrected calls per imagined future); the norm weights are attached once via `add_reward_function` in the simulation setup [Code, `src/utis/simulation.py`]. Oncoming's norm set additionally treats speed deviation as non-compliance (a braking oncoming vehicle is "abnormal"), chapter 07 [Code].

Chapter 7: normative driving — what defines "normal", and every knob that moves it

"Normal" appears twice, and they are different objects

The model contains two separate definitions of normal driving, implemented in different places, doing different jobs:

- **The driver's own normal** — the preference prior: how *my* drive is supposed to go. This shapes what the driver does.
- **Normal for the others** — the norm geometry of chapter 06: what *that other vehicle* is expected to do. This shapes what the driver predicts, hence when it worries.

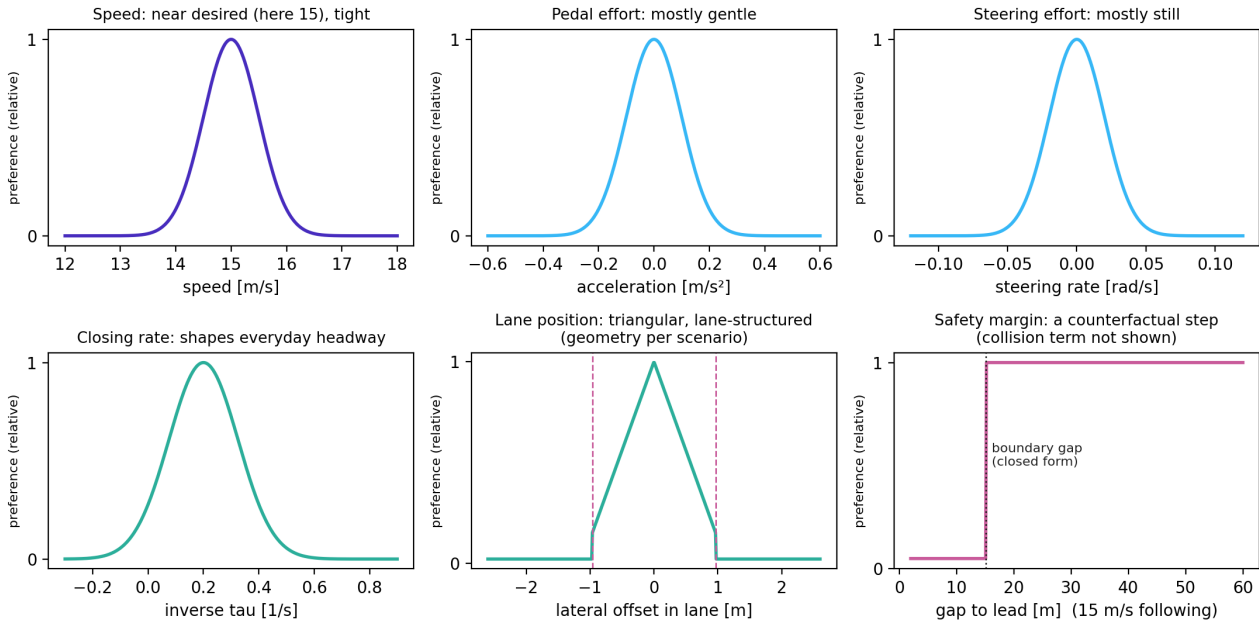
We take them in turn.

Part A: the driver's own normal — six preference terms

The preference prior is a product of six independent terms [SI §2.4]. Each term is a distribution over one observed quantity: it says which values are treated as unremarkable and how quickly departures become objectionable. Because the terms multiply, the model's "how the drive should go" is simply the six read together — and because they are independent, every exceedance can be attributed to the term responsible.

Term	Plain meaning	The knobs	Turning them does
Speed	I intend to travel near my desired speed	desired speed; tolerance (sd 0.5 m/s)	tighter tolerance → speed held more rigidly, stronger urge to return to it after braking
Pedal effort	acceleration should be mostly gentle	tolerance (sd 0.1 m/s ²)	smaller → smoother driving, later/harder emergency trade-off felt
Steering effort	the wheel should be mostly still	tolerance (sd 0.02 rad/s)	smaller → steering escapes score worse, braking favored
Lane position	stay centered in a real lane	lane geometry; lane-change cost; road-leave cost	the scenario-shaped term — see below
Closing rate (inverse-tau)	do not close on the vehicle ahead faster than a TTC of about 5 s	preferred inverse-tau level and width	one-sided as released: closing slower than that, holding the gap, or falling back costs nothing, so this term bounds the <i>approach rate</i> and does not shape the following distance itself [Code: <code>reward.py</code>]
Collision & safety margin	collisions are unacceptable, scaled by severity — and so are states from which only heroic braking would save me	collision cost; severity floor; assumed worst-case lead braking; assumed own reaction time (1 s)	the safety-margin term — see below

The six preference terms — the driver's own 'normal', term by term (shapes to scale; parameter values from the shipped configuration)



Three of the six deserve a longer look.

The lane term is where scenarios differ. "Centered in my lane" requires knowing what lanes exist and which direction they serve; that geometry is hand-drawn per scenario [Code]. Rear-end's road adds explicit costs on dawdling between lanes or aborting a lane change — hand-built craftsmanship, not derived theory, and worth knowing about before attributing its effects to deep principles [Code] [Opinion]. There is no map format: a new road means drawing new geometry (chapter 04, checklist item 4).

The closing-rate term defines everyday tailgating comfort. It is easy to overlook — the article barely mentions it — but without it the model would happily sit at a tiny, technically-safe gap. It encodes "being close and closing feels wrong before it is dangerous" — which, the way we read it, is a comfort standard distinct from the safety margin [SI] [Opinion].

The safety-margin term is a counterfactual, and its assumptions are the boundary's location. It scores the present state by a what-if: *if* the lead braked at an assumed worst-case level *and* I responded only after an assumed reaction time, would ordinary braking still suffice? Both assumptions are parameters — the assumed worst case is calibrated per scenario [SI] — and any absolute number derived from this term (a critical headway, a threshold gap) inherits them and should be quoted with them.

Added 2026-09-03. Two properties of the released form matter to anyone taking this term outside the published scenarios, and we have met both. The term is an **indicator**, not a graded quantity: when the required deceleration exceeds what the vehicle can do (8 m/s^2), or the gap has already closed before the reaction time ends, a fixed penalty of half the collision cost at the severity floor is charged, and otherwise nothing; and it applies only while the other vehicle is ahead and within 1.15 vehicle widths laterally, the same box the collision test uses [Code: `reward.py`, `decoder.py`]. So a vehicle that is *entering* the lane — a highway cut-in spends about 2.5 s straddling the boundary in recorded clips — is invisible to the term until the box test flips, and the term then switches on at full strength: at 30 m/s the indicator is already violated at a 10 m gap and at a 21 m gap alike, so two clips of very different urgency receive the same score [Study]. When we instead rendered the same counterfactual continuously, as the relative speed that would remain at impact under full braking after the reaction time, and asked whether it orders people's intervention judgments on those clips, it did not. In that study's regime (ego at 110–130 km/h, gaps 4–80 m) the counterfactual is violated almost everywhere, so the quantity measures how bad the worst case would be, which the absolute speeds set far more than the current gap does: about 0.3 m/s per meter of gap against 1.5–2 m/s per m/s of either vehicle's speed. Within rows of matched time-to-collision it ranks clips by how fast the vehicles are going, while raters rank them by how far apart they are, and a plain gap threshold predicts the judgments far better: held-out RMSE 0.15 against 0.35 for our continuous rendering of the six-term preference deficit, with chance at 0.32. About half of that deficit's loss is attributable to a lane-entry gate of our own construction, not the released box test, and the rest to the counterfactual magnitude; with our gate removed the deficit still scores 0.26 [Study]. We read this narrowly. A

worst-case stopping counterfactual may be exactly the right reference for collision avoidance, where the outcome does depend on where the target will be and how fast everyone is going, and the wrong one for the earlier, comfort-like judgment the raters were making, which on these data looks like a distance judgment at a given closing rate; chapter 03 reports that the model's own looming variable is the scalar that fits those judgments best [Opinion].

New in this version, 28 September 2026: not in the edition you received. How often the terms act on real traffic. Evaluated on about a million car-following samples from the highD dataset, the calibrated safety-margin term is violated in 1.7% of steady following and the inverse-tau term in 0.05%. The headway at which the margin starts to cost, at equal speeds, lies at or below the fifth percentile of real headways in four of five speed bands (0.70 s at 40–70 km/h down to 0.05 s above 130 km/h, against real fifth percentiles of 0.54–0.67 s). On 2,208 real closing cut-ins, at the moment the cutting-in car enters the lane, 99.5% have a time to collision of 5 s or more, where the one-sided inverse-tau term sits at its floor and costs nothing; real followers nevertheless brake in some of those cases, mostly gently. Both terms are shaped for avoiding collisions, which is what the model is for, and leave most of ordinary driving to the gentler terms [Study].

Part B: normal for the others — the three scenarios in detail

The norm geometry that chapter 06's prediction machinery consumes, read directly from the code [Code]:

- **Rear-end.** Normal = the lead staying within the lane's width. Graded: fully normal in-lane; a small factor when marginally outside; a much smaller factor further out. Nothing about speed — a lead may brake without becoming "abnormal", which is consistent with braking leads being the scenario's whole point.
- **Oncoming.** Normal = staying in *its own* lane, **and** holding its speed: the compliance weight falls off quadratically as its speed departs from the nominal one. An oncoming vehicle that brakes hard is treated as abnormal even before it crosses the line — the earliest warning the driver can get in this geometry.
- **Intersection.** Normal = respecting the junction's geometry: not entering our carriageway past the yield line ("ignoring the light"), not cutting the corner arc, not leaving the paved area. Drawn as literal regions of the junction plan.

The pattern worth naming: each scenario's norm set encodes *the specific way that scenario's threat announces itself* — lane exit for oncoming, junction entry for crossing. The way we read it, writing a new scenario's norms (checklist item 5) amounts to answering: "what is the earliest observable sign, in this geometry, that the other agent has stopped being ordinary?" [Opinion]

Part C: how normal can be changed

Because both normals are explicit objects, changing them is parameter work, not rearchitecting — with effects that are predictable in direction.

Traits (stable differences between drivers). A cautious driver: the assumed worst-case lead braking is made more severe (they plan for worse) and the pedal tolerance tightened. A hurried driver: higher desired speed, tighter speed tolerance, shorter assumed reaction time. A smooth-ride chauffeur: pedal and steering tolerances tightened, closing-rate preference widened. Each is a handful of interpretable numbers [Opinion, though the parameter meanings are the paper's].

States (the same driver on a bad day). Temporary motives — hurry, anger, social pressure — become *temporary reshapings of the preference prior*: a shortened assumed reaction time, a relaxed safety assumption, a higher desired speed. Because the safety-margin term is a closed-form counterfactual, the behavioral consequences of such reshapings are computable in advance rather than only simulable, which makes them falsifiable predictions about how accepted margins shift under manipulated motives [Opinion].

Norm changes (the others' normal). Widening the lead's normal region makes the driver slower to worry; tightening it makes the driver jumpy. A learned, data-driven norm set — replacing the hand geometry with distributions fitted to observed traffic — is the extension we would rank most valuable [Opinion].

Part D: what, ultimately, defines it

Honesty about provenance: the shape of every preference term is **specified by the authors**, not derived from first principles; thirteen parameters were hand-tuned to produce human-like behavior [Paper]; one (the assumed worst-case braking) is calibrated against a separate free-following dataset per scenario [SI]; none are fitted to the

conflict data they are evaluated on. So "normative driving" in this model is a *stated hypothesis about drivers' standards*, made falsifiable by its behavioral consequences — response times, maneuver choices, headways — rather than an empirical measurement of those standards.

Notes for the mathematically curious

Level 1 — preference as log-probability. Each term contributes a log-probability; the six add. "Zero cost" is the mode of each distribution; the cost of a departure is the log-density drop. The additive decomposition is what lets an exceedance be blamed on a term: the residual ϵ (chapter 02) is a sum of per-term residuals.

Level 2 — forms and numbers. Speed, pedal, steering: Gaussians with sds 0.5 m/s, 0.1 m/s², 0.02 rad/s around desired speed / 0 / 0 — with two code-defined details on the pedal term: positive accelerations are doubled before the Gaussian, and the quantity penalized is the total acceleration $\sqrt{a_{\text{lat}}^2 + a_{\text{long}}^2}$, not the longitudinal component [Code: `reward.py`]. Lane: triangular within-lane density with hard log-costs −1000 (lane boundary) and −15000 (road edge; the value in the released code), lane-structured per scenario (SI Eq. 52). Inverse-tau: Gaussian on $1/\tau$ with mean 0.2 s^{−1}, sd 0.125 s^{−1}, evaluated on $\max(1/\tau, 0.2)$ so that it is one-sided in the released code [Code: `reward.py`; SI Eq. 48 writes the symmetric form]. Collision: cost −10000 times a severity factor $0.2 + 0.8 \cdot \Delta v / (10 \text{ m/s})$, with Δv the absolute longitudinal speed difference at first contact, so 0.2 for a touch, 1.0 at 10 m/s and, uncapped, 1.8 at 20 m/s — the 0.2 floor is SI Eq. 48, not a fudge [Code: `reward.py`; revised 2026-09-03: the first edition wrote $\max(\Delta v / 10, 0.2)$, which is not the released form]. Safety margin (SI Eqs. 49–51): required deceleration under the counterfactual (lead brakes at $\min(\text{observed}, \text{assumed worst})$; own response after $t_{\text{react}} = 1 \text{ s}$), compared against the achievable 8 m/s² (a_{max}); charged as an indicator, $0.5 \times \text{collision cost} \times \text{the severity factor at its floor}$, that is −1000 per step, whenever the required deceleration exceeds 8 m/s² or the gap closes before the reaction time ends, and 0 otherwise. Both this term and the inverse-tau term are multiplied by the lateral applicability test (target ahead, within 1.15 widths) [Code: `reward.py`, `decoder.py::test_looming_viability`]. Norm weights: Part B's geometries with factors 0.001 and 0.001×0.01 (`weigh_particles`, `full_violation_factor`); oncoming's speed compliance $1 - 2.25 (v/v_0 - 1)^2$, clipped [Code]. The thirteen hand-tuned parameters are listed in the paper's methods [Paper].

Chapter 8: the gaze system, and what impairment-related machinery exists today

What "crash causation" needs from a driver model

The mechanisms behind real crashes — in the human-factors reading — are rarely exotic: eyes off the road at the wrong moment, expectations that the situation then violates, impaired or slowed responses, and degraded perception. A driver model useful for causation research must be able to *produce* crashes through these mechanisms, not just fail randomly. The answer to "can this model do that today" is: partly — and more than the collision-avoidance paper itself advertises, because much of the machinery ships in the code in dormant form.

What exists in the released code today

1. A complete, dormant gaze system [Code]. The common code — not a fork, the code that ran the published results — carries a two-state gaze variable (eyes on road / off road) threaded through the whole architecture:

- the *dynamics* include a gaze state with switching probabilities between on and off;
- the *observation model* multiplies perception noise by a factor (3 in the shipped configuration) when gaze is off road — looking away does not blind the driver, it degrades evidence quality by a set ratio;
- the *belief machinery* reserves the first two state dimensions for gaze (visible in the OSF belief arrays, where they sit constant [OSF]);
- the *preference vocabulary* includes a gaze term (`road_gaze_preference`, set to zero in all published collision-avoidance runs [OSF]);
- and the *planner* contains the line that turns it all off: a hard-coded "avoid off gaze right now" [Code, `mpc_discrete.py`].

In other words: off-road glances are implemented as *actions the driver could choose*, with an evidence price attached, and the published collision-avoidance model was simply forbidden from choosing them. The earlier paper in this line (Engström et al. 2024) demonstrates exactly this machinery on uncertainty-and-looking tasks [Paper]; the

collision-avoidance paper switched it off to isolate avoidance behavior.

Added 2026-09-03. We have since exercised the dormant gate by forcing off-road glance schedules on the released rear-end model, with the planner's prohibition lifted for the schedule and everything else as shipped. The result is worth knowing before using the gate for causation work: **it blocks new observations, not inference**. A driver that has registered the lead's braking and is then blinded keeps responding *during* the glance, at essentially the attentive onset, even under a near-total observation blackout, because the belief cloud coasts forward on its own norm-shaped prediction (chapter 06) and the accumulator keeps filling from the extrapolated evidence. The architecture therefore predicts that a glance beginning *after* the conflict has been registered costs little, while a glance covering the onset costs the whole detection; a model that assumes no accumulation while the eyes are off the road predicts the opposite in the first case. That is a testable behavioral distinction that neither description states [Study].

2. Perception-quality causation [Code] [SI]. Observation noise scales are parameters, and looming makes perceptual difficulty state-dependent for free: small visual angles and low expansion rates are genuinely harder. Fog, darkness, or a small motorcycle are representable today as noise-scale and geometry changes — with the detection threshold producing late detection *mechanistically* rather than by adding delay.

3. Expectation-based causation [Code] [Paper]. Chapter 06's norm trust is a looked-but-did-not-expect mechanism already: a driver whose norms say "oncoming vehicles stay in their lane" allocates prediction weight accordingly, and is structurally late for the rare violator. Mis-calibrated trust — the classic expectancy crash — is a parameter setting, not a new mechanism.

4. A response-vigor dial [Paper]. The evidence-accumulation rate λ globally scales how fast surprise converts into action. It is the model's single most response-time-sensitive parameter, which makes it a blunt but honest "generalized impairment" knob.

What does not exist

No fatigue or drowsiness dynamics (nothing varies with time-on-task; no microsleep process). No cognitive load or dual-task interference beyond the gaze dichotomy. No alcohol/drug pharmacodynamics. No individual-differences layer — one parameter set per run. No learning: expectations do not drift with exposure. These absences are real; how much of each can be composed from the parts above is, in our view, one of the more interesting open questions this architecture poses [Opinion].

Notes for the mathematically curious

Level 1 — why the gaze system prices glances correctly. With gaze off road, observation noise is multiplied by a factor ≥ 1 , so the observation likelihood flattens and the belief cloud's spread grows; epistemic value (expected uncertainty reduction) of returning gaze to the road grows with it. A glance therefore carries an automatically increasing price as the situation's uncertainty compounds — the model looks back *because it is worth it*, reproducing the uncertainty-driven glance rhythm of the 2024 paper [Paper].

Level 2 — the implementation sites. Gaze state: first two dimensions of the belief state (`b[... , :2]` in the OSF arrays, constant in the published runs [OSF]); transition probability and off-gaze noise factor: `src/common/dynamics.py` (parameter `p`, `I_factor`) and `src/common/decoder.py` (noise multiplication under off gaze); the hard-coded prohibition: `src/common/mpc_discrete.py` ("Avoid off gaze right now (hardcoded)"); the preference slot: `road_gaze_preference` in every `Setups_*.xlsx`, value 0 [OSF].

Chapter 9: modifying the model, and knowing whether the modification is right

The seams

The previous chapters located every place the model is *meant* to be changed. Collected in one table:

Seam	What it represents	Where	Chapter
Preference knobs	the driver's own standards; traits and states	<code>reward.py</code> params / Setups columns	07
Safety-margin assumptions	worst case planned for; reaction time budgeted	preference term + calibration table	07, 04

Seam	What it represents	Where	Chapter
Norm geometry	what others are expected to do	<code>reward.py::get_weights</code>	06, 07
Assumed other-agent variability	what kind of agent I am facing	<code>a_sd_model, w_sd_model</code>	04, 06
Perception noise & looming	visibility, conspicuity, sensory quality	decoder noise scales, threshold	03, 08
Accumulation rate λ	evidence-to-action vigor	<code>EA_fac</code>	05, 08
Gaze system	attention on or off the road	dormant; chapter 08	08
Scenario script	the world and its threat	<code>dynamics_true.py</code>	04

A modification that does not fit one of these seams — one that needs the planner rewritten or a new state variable threaded through the belief machinery — is a research project, not a modification, and should be costed accordingly [Opinion].

This chapter is about *what* to change and how to know the change behaves; the companion question — how the new parameter values themselves should be obtained, and the identifiability traps in fitting them — is chapter 10.

The validation ladder

The question "how would we validate a change" has, in our view, one good general answer: climb, and do not skip rungs [Opinion]. Each rung has a cheap failure mode that the rung above cannot detect.

Rung 0 — property checks (minutes). Verify the changed component against its own specification, ideally by two independent routes (a closed form against a numeric evaluation; a reimplementation against the original). Sign errors and unit slips produce plausible-looking wrong numbers that eyeballing does not catch; two independent routes do.

Rung 1 — mechanism check (hours, static). Confirm the proximal effect: the knob you turned moves the quantity it is supposed to move, in the right direction, by roughly the expected amount, with other quantities still. For preference changes this needs no simulation at all — the preference function can be evaluated pointwise on recorded or constructed states.

Rung 2 — the ablation discipline (days of CPU, or free from the deposit). Show what the mechanism *contributes* by removing it. The paper's own Figure 6 does this for seven mechanisms, and — the practical gift — the OSF deposit contains the complete simulation output for every one of those ablations, seven variants $\times$ the full rear-end grid, already run [OSF]. Before running anything: check whether the ablation you need is already in `Setups_rear_end.xlsx` (no evidence accumulation; no prediction noise; no pedal constraints; no looming perception; no looming threshold; no norm conditioning; no epistemic value). A new mechanism should ship with its own ablation the same way: the model with and without it, same grid, same seeds.

Rung 3 — distribution comparison against the reference (free, thanks to the deposit). Any closed-loop change must reproduce the *unchanged* behaviors: response-time distributions, maneuver mix, collision rates, per condition, against the authors' own output — not against numbers read off figures. A modified model earns trust by matching the baseline where it should match and departing only where its mechanism says it should depart.

Rung 4 — human data. Only rungs 0–3 make rung 4 interpretable: when the modified model meets human data (response-time curves, glance statistics), any mismatch can be attributed to the mechanism under test rather than to a broken foundation.

Practical constraints that shape all of this

- **The closed loop is expensive.** The code is written for GPU; on CPU, a single scenario is hours and a condition grid is a cluster job. Consequences: prefer rung 1's static evaluation wherever the question allows, and mine the deposit before simulating anything — rungs 2–3 are often free.
- **Keep the reference implementation clean.** Modifications belong in forks, never silently in a local copy of the released code, so that "the reference model" stays a fixed point all comparisons share.
- **Record complete configurations.** Every result should carry its full parameter set the way the authors' Setups tables do; a result whose configuration is lost cannot be compared with anything.

Notes for the mathematically curious

Level 1 — why rung 1 is often static. Preference changes alter the pointwise landscape over states; their first-order behavioral consequences (threshold locations, term attributions) are properties of that landscape and need no dynamics. Only changes that alter *timing* (λ , perception noise, gaze) or *prediction* (norms, variability dials) require the loop.

Level 2 — the ablation columns. In every `Setups_*.xlsx`: `EA_mode = None` (no evidence accumulation — the model re-plans continuously), `noise_pred_fac = 0.002` (no prediction noise), `use_pedals = 0`, `use_looming_perception = 0`, `looming_threshold = 0`, `N_norm = 1` (no norm conditioning), `alpha = 0` (no epistemic value) [OSF]. The deposit holds $7 \times 28 = 196$ ablation runs for rear-end alongside the 28 baseline ones; oncoming and intersection carry the same structure at their grid sizes.

Chapter 10: calibration and parameter fitting — how the numbers get their values

Three ways a number gets into the model — fitting is only one of them

A natural first assumption is that a model like this is "fitted to data" the way a regression is. It is not, and the distinction matters for anyone planning to change it. Every number in the model arrived by one of three routes, and each route carries its own obligations when you replace the number:

- 1 **Specified** — taken from physics, geometry, or an independent literature. Vehicle dimensions, maximum braking, the timestep; the looming detection threshold comes from the psychophysics literature, not from fitting [SI]. Obligation when changing: cite the independent source, or admit the number has become a fitted one.
- 2 **Calibrated on normal driving** — set so that *everyday* behavior comes out right, then left alone when conflicts are simulated. The flagship example is below. Obligation: the calibration data must be separate from the evaluation data, and the calibration must *cover* the conditions you will run.
- 3 **Tuned** — thirteen parameters were adjusted by hand until behavior looked human at the qualitative level [Paper]. Honest, but the route with the least protection against fooling yourself. Obligation: whatever was tuned must face validation it was not tuned toward — which is exactly the role of the held-out intersection scenario.

The calibration exemplar, worth copying: the free-following lookup

The single most consequential preference parameter — the assumed worst-case deceleration of the other vehicle, which sets where the safety margin sits (chapter 07) — is neither specified nor tuned. It is calibrated, and the mechanism deserves to be understood because it is the pattern any extension should copy [Code, `find_parameters` in `src/utils/simulation.py`; SI]:

- 1 Run (once) a large grid of *free-following* simulations — no conflict, just steady car-following — across speeds, accumulator rates, and candidate worst-case assumptions, recording the steady headway each combination settles into.
- 2 Invert the table: given a speed and a desired everyday headway, look up (with interpolation) the worst-case assumption that *produces* it.
- 3 Use that value, unchanged, when simulating conflicts.

The logic is quietly elegant: the paranoia parameter is disciplined by ordinary behavior. A driver who assumed worse would follow further back *all the time*; observed comfortable headways therefore pin down the assumption without touching any conflict data. Calibrate on the quiet regime, predict the loud one — the same separation chapter 05 describes behaviorally, used as an inference principle.

One general lesson attaches to this exemplar: **a calibration is only as good as its coverage**. The shipped lookup table spans steady-state headways up to about 1.0–2.1 s depending on speed; outside that range the interpolation clamps to the table edge and the parameter saturates at its most pessimistic value, -8 m/s^2 [Code]. Coverage failures do not announce themselves — the model still runs — so checking that a calibration table brackets every condition one intends to simulate is a one-line assertion worth writing.

Identifiability: the central danger of fitting this model

The model's parameters do not map one-to-one onto observables. Several knobs move the same behavioral output, which means a good fit does not tell you which knob was responsible — and a fitted value can be badly wrong while the fit looks fine:

Observable	Moved by (at least)
Response time	accumulator rate λ , looming threshold, perception noise, prediction noise, norm trust
Accepted headway / margin location	assumed worst-case braking and reaction-time budget (chapter 07)
Maneuver choice vs speed	steering-effort tolerance, lane costs, collision severity scaling
"Cautiousness" overall	collision cost, severity floor, safety-margin assumptions — jointly

One pair deserves its own sentence: the accumulation rate λ and the collision cost g_C are a single degree of freedom for response timing. The accumulator fires when $\lambda \cdot \Sigma \epsilon$ passes 1, and in a conflict ϵ is dominated by terms proportional to g_C , so only their product sets the timing; g_C is separately identified through the maneuver trade-offs, λ is not [Opinion].

The defenses are standard but non-optional:

- **Fix everything you can by routes 1 and 2** before fitting anything by route 3; every parameter removed from the fit is a confound removed from the interpretation.
- **Fit the smallest subset that your hypothesis is about**, holding the rest at reference values — and say so.
- **Probe the objective around the optimum.** If a parameter can move substantially with little cost to the fit, report the range, not the point; a flat direction is a finding about the model, not an inconvenience.
- **Prefer observables that separate the knobs.** Response time alone cannot distinguish λ from perception noise; response time *as a function of condition* (urgency, speed, visibility) begins to, because the knobs bend that curve differently. Distributions beat means for the same reason.

The mechanics: how fitting actually proceeds here

There is no formula linking parameters to data — the model is a simulator, so fitting means comparing simulated and observed behavior and searching. Three practical consequences [Paper] [OSF]:

- **Fit summaries, not trajectories.** The paper's own comparisons use response-time distributions per condition, maneuver-choice proportions, and deceleration profiles, with distribution distances (Jensen–Shannon, Wasserstein) and regression-based error bands as the yardsticks. Any refit should use the same summaries first, so results stay comparable.
- **Respect the noise floor.** Each condition was run with 32 random seeds, and seed-to-seed spread is substantial; a fit that chases differences smaller than the seed spread is fitting noise. Simulate enough seeds to know the model's own variability before crediting a parameter change with an improvement.
- **Be static wherever possible.** Preference parameters can be evaluated against recorded kinematics without running the loop; only timing-and-prediction parameters (λ , noise scales, norm trust) genuinely require simulation — and for those, the deposit's precomputed grids [OSF] and coarse-to-fine searches make the difference between a week and a year.

Dos and don'ts, collected

Do

- Fix by physics and literature first; calibrate on normal driving second; fit last, and least.
- Check calibration coverage against every condition you will run, mechanically.
- Fit distributions per condition, never grand means; simulate enough seeds to know the model's own spread first.
- Keep the fitted-on data and the validated-on data disjoint, and keep one scenario untouched end to end — the held-out discipline is the model line's own best habit.
- Re-run the ablation for any mechanism whose parameters you refit (chapter 09, rung 2): a refit can quietly transfer a job from one mechanism to another.
- Record complete parameter sets with every result, the way the Setups tables do — including which route (specified / calibrated / tuned / fitted) each value came by.
- Report flat directions and ranges; treat weak identification as a finding.

Don't

- Don't hand-tune against the data you will call validation — with this many degrees of freedom the fit will succeed and mean nothing.
- Don't fit λ and perception noise jointly from response times alone; they are not separable there (use condition structure, or fix one independently).
- Don't quote absolute margin or headway values without the assumptions they inherit (chapter 07).
- Don't let calibration tables or fitted values live outside version control and the parameter record; a number whose provenance is lost reverts to "tuned".

Notes for the mathematically curious

Level 1 — the fitting problem stated. With simulator $M(\theta)$ mapping parameters to behavior distributions and observed summaries s_{obs} , fitting is $\text{argmin}_{\theta} D(s(M(\theta)), s_{\text{obs}})$ for a chosen distance D over chosen summaries s — a simulation-based (likelihood-free) inference problem. Identifiability is the geometry of D around the optimum: flat valleys are unidentified parameter combinations. The static shortcut works whenever s depends on θ only through the preference function evaluated on fixed kinematics — then the simulator drops out and the search is direct.

Level 2 — specifics. The free-following lookup: `find_parameters(v_tar, EA_fac, noise_pred_fac, H, d_phi_thres, THW_des)` multi-linearly interpolates a pre-run grid (`Analysis_following.xlsx`) to return the `a_tar_min` (and residual speed offset) consistent with the desired steady headway; the returned value is clamped at the grid edge — the coverage property above [Code]. The paper's comparison metrics: MAE with bootstrapped uncertainty via Bayesian linear regression on residuals, Jensen–Shannon divergence for categorical outcomes, Wasserstein distance for response-time distributions, with a signal-to-noise criterion for reporting [Paper].

Chapter 11: the code and data map — from concept to file, class, and parameter

*Paths are relative to the root of the authors' released repository [Code]. Parameter names are the columns of the OSF 'Setups.xlsx' tables, which double as the complete configuration record of every published run [OSF].**

The reference implementation, oriented in one table

Top level: one `simulation_<scenario>.py` runner and one `Analysis_<scenario>.py` per scenario (plus `_SA` sensitivity variants); the runners assemble a configuration dictionary, call the shared machinery, and write the pickle/xlsx outputs mirrored in the OSF deposit.

Concept (chapter)	File in <code>src/common/</code>	What to look for
World physics	<code>bicycle.py</code> , <code>environment.py</code>	bicycle model both vehicles share
Looming senses (03)	<code>decoder.py</code> , <code>encoder.py</code>	observation construction; the looming transform; the off-gaze noise factor <code>I_factor</code>
Belief cloud (06)	<code>particle_filter.py</code> , <code>kde.py</code> , <code>distributions.py</code>	weighting, resampling; the KDE/mixture belief representation
Imagined futures + norm trust (06)	<code>dynamics.py</code>	<code>forward_tar_agent</code> , <code>normative_probability</code> ; the <code>N_norm/H_norm</code> sampling and the trust cap
Scoring, pragmatic + epistemic (03)	<code>belief_reward.py</code>	expected-free-energy assembly over particles
The planner and the surprise gate (03, 05)	<code>mpc_discrete.py</code>	CEM loop; evidence accumulation; the hard-coded "avoid off gaze" line (08)
Gaze dynamics (08)	<code>dynamics.py</code> (gaze block)	two-state gaze, switching probability <code>p</code>
Run orchestration	<code>src/utls/simulation.py</code> , <code>saving.py</code>	how a config becomes a run becomes a pickle

Per scenario (chapter 04): `src/<scenario>/dynamics_true.py` (the world and the other vehicle's script), `decoder_true.py` (true-state $\rightarrow$ observations), `reward.py` (the preference terms with the scenario's lane geometry,

and `get_weights` — the norm geometry of chapters 06–07). The rear-end scenario used for the published grid lives in `src/rear_end_test/`.

The parameters that matter most, by name

All 65 configuration columns are in every `Setups_*.xlsx`; these are the ones this handbook's chapters keep returning to:

Column	Plain meaning	Chapter
<code>v_ego_sd_des</code> , <code>a_ego_sd_des</code> , <code>w_ego_sd_des</code>	tolerances of the speed / pedal / steering preferences	07
<code>lane_change_cost</code> , <code>road_leave_cost</code> , <code>collision_cost</code>	the hard costs	07
<code>a_sd_model</code> , <code>w_sd_model</code>	assumed other-agent variability (the scenario-type dial)	04, 06
<code>N_norm</code> , <code>H_norm</code> , <code>weigh_particles</code> , <code>full_violation_factor</code>	norm-conditioning strength and geometry factors	06
<code>alpha</code>	weight of epistemic value	03
<code>EA_mode</code> , <code>EA_fac</code> , <code>EA_init</code>	evidence accumulation on/off, rate λ , starting level	02, 05
<code>use_looming_perception</code> , <code>looming_threshold</code>	the visual channel and its detection floor	03, 08
<code>x_sd_perc ... w_sd_perc</code>	perception noise scales	03, 08
<code>num_plans</code> , <code>H</code>	planning budget and horizon	03
<code>road_gaze_preference</code> , <code>x_ego...v_tar</code> , <code>T</code>	gaze preference (0 in all published runs); stage-setting	08, 04
ablation columns	the seven Figure-6 switches	09

The data that goes with the code

The OSF deposit is the third leg of the release: per-run configuration tables (`Setups_*.xlsx`), outcome summaries (`Analysis_*.xlsx`), and per-timestep pickles (true states `e`, observations `o`, beliefs `b` with weights `w`, planned and reference policies `a_cont` / `a_cont_init`, pragmatic-value components `v`) for every published run of all three scenarios, ablations included. One practical note for new users of the deposit: the policy arrays are indexed (horizon, timesteps, seeds, 2); the shape of `e` disambiguates when in doubt [OSF].

Two first exercises

For a new reader's first afternoon:

- Meet the model's output.** Load one deposit pickle (`Results_rear_end/Exp_7/Exp_7.pkl`), plot speed, gap, and executed acceleration for a few seeds. You are reproducing chapter 02's figure; the axis-order note above is the only trap.
- Experience the loop.** Run one short rear-end condition through `simulation_rear_end.py`'s `simulate()` path for a single initial condition (a small wrapper that skips the full sweep is a half-hour of work, and worth writing with a checkpoint that saves partial results as it goes — the loop is slow on CPU).

Chapter 12: glossary — one idea, three vocabularies

The middle columns are our translations; where a mapping is loose we say so rather than force it.

The Rosetta stone

Term here	Engineering / ML reading	Human-factors reading
Generative model	internal simulator / world model	the driver's understanding of how traffic works
Belief (particle cloud)	posterior state estimate with honest uncertainty	situation awareness, held with degrees of confidence
Observation	sensor reading	what perception currently delivers
Looming	angular-size channel with state-dependent noise	optical expansion, the classic visual cue
Surprise	negative log-likelihood of what happened	expectancy violation

Term here	Engineering / ML reading	Human-factors reading
Free energy	a computable bound on model misfit	(no native equivalent — "how badly my picture of the situation fits")
Preference prior	goal specification, written as a distribution	motivation; how the drive is supposed to go
Pragmatic value	expected goal achievement of a plan	progress and safety satisfaction
Epistemic value	expected information gain of a plan	the pull to look, probe, and resolve uncertainty
Policy	planned control sequence	intended maneuver
Expected free energy	plan cost = pragmatic + epistemic in one currency	the felt overall "rightness" of an intended course
Bounded planning (CEM)	sampled, budgeted trajectory optimization	satisficing; good-enough decision making
Norm (about others)	prior over other agents' trajectories, geometry-shaped	expectancy about other road users' behavior
Norm-conditioning trust cap	prior weight gated by observed compliance	trust extended while earned, withdrawn on evidence
Evidence accumulation (E, λ)	leaky-free integrator to threshold	the response-timing process of accumulator models
Residual information (ϵ)	shortfall of current plan vs best achievable, in nats	how far the situation has left "as it should be"
Precision	inverse variance; confidence weighting	how much a cue or expectation is trusted
Ablation	mechanism knocked out, behavior compared	showing a mechanism matters by removing it
Calibration (vs fitting)	setting a parameter from separate, non-evaluation data	grounding a number in ordinary behavior before predicting rare events
Validation (vs fitting)	testing against data no parameter ever saw	the held-out discipline
Identifiability	whether data can pin a parameter down uniquely	whether two explanations of the same behavior can be told apart
Summary statistic	the condensed observable a simulator fit targets	the behavioral measure (RT distribution, maneuver share) standing in for raw data

False friends — words that do not mean what they usually mean

- **Surprise** is a *quantity*, not an emotion. A state can carry surprise the driver would never report feeling; the model's "surprise" is closer to *mismatch*.
- **Preference** is not a choice or a ranking; it is a probability distribution stating which futures are treated as unremarkable. Wanting and expecting are deliberately the same object.
- **Reward** appears in the code (`reward.py`) but is *not* RL reward: nothing is being maximized by trial-and-error learning. The file computes log-preference.
- **Norm** is not a traffic rule. It is a description of what other agents typically do, used for prediction — a violated norm is information, not an offense.
- **Free energy** has no thermodynamic content whatsoever (chapter 01).
- **Belief** carries no conscious commitment — it is a weighted hypothesis set.
- **Optimal** almost never applies: the planner is deliberately budgeted, and the model's humanlikeness partly *depends* on its suboptimality (chapter 03).
- **Agent** means the simulated driver — but in `dynamics_true.py` "target agent" is a scripted puppet with no agency at all (chapter 06).
- **Epistemic** does not mean abstract knowledge-seeking; operationally it is "this plan will let me see better".

Frequently confused — short answers

Is this reinforcement learning? No. Nothing is learned from reward across episodes; there is no training loop. The preferences are specified, the behavior is computed fresh each run. The resemblance is only that both talk about value.

Is it optimal control with extra words? Closer, but two differences do real work: the cost function is a probability distribution (which is what lets the same object define surprise, and hence timing), and the planner is deliberately bounded (which is where the human character of the maneuvers comes from). An optimal-control reading also has no native account of the epistemic term.

If the agent minimizes surprise, why doesn't it park in a dark garage? The famous "dark-room" objection.

Because surprise is measured against the *preference prior*, and the preference prior of a driver says "I am making progress at my desired speed". Sitting still is maximally surprising to an agent whose expected world involves getting somewhere. (Chapter 13 covers the debate around this answer.)

Does the model want to be surprised (curiosity) or not (comfort)? Both, coherently: it avoids *pragmatic* surprise (departures from the preferred future) while seeking observations that reduce uncertainty — the epistemic term. The two are added in one currency, which is the framework's central accounting trick (chapter 03).

Is the particle filter what makes the other car move? No — a common confusion. The other car follows a script (chapter 06). The particle filter is the *driver's uncertainty* about the world, including about that scripted car.

Are the parameters fitted to the crash data? Thirteen were hand-tuned; the assumed worst-case braking is calibrated on separate free-following data; the intersection scenario was held out entirely [Paper]. No parameter was fitted to the conflict responses the model is evaluated on.

Do I have to believe the brain minimizes free energy? No. Chapter 01's permission slip: the model stands or falls as an empirical driver model, whatever the grand theory's fate.

Noise or uncertainty — which is which? Noise is in the world and the senses (parameters); uncertainty is in the beliefs (the cloud's spread, computed). Turning noise up raises uncertainty, but uncertainty also rises with distance, occlusion, and gaze — that is the point of carrying it explicitly.

Chapter 13: appendix — the deep end

This appendix exists so that the main chapters could stay out of these waters. Nothing here is needed to use the model; everything here is where a curious reader should go next. It is a guide with commentary, not a tutorial — each section says what the thing is, why the main text could skip it, and where the primary treatment lives.

A. The free-energy principle proper

What it is. The claim beneath the framework: any system that maintains itself against dispersion — a cell, a brain, arguably an organism-environment loop — can be described as *if* it minimizes variational free energy, because persisting just is keeping sensory states within expected bounds. In its strongest form the claim is presented as close to tautological: things that exist are things that model their environment well enough to keep existing.

Why the main text skipped it. The driver model uses free energy as an engineering objective (chapter 01, claim 3). Whether the principle is a deep truth, a useful reformulation, or an unfalsifiable framing is irrelevant to whether the model predicts human braking — and the debate is long.

Where to go. Friston, "The free-energy principle: a unified brain theory?" (Nature Reviews Neuroscience, 2010) — the canonical statement. Friston, "A free energy principle for a particular physics" (2019 preprint) — the strongest, most mathematical form. For the driving-adjacent reading, the introduction of Engström et al. (2024) states exactly how much of the principle the model line actually uses.

B. Variational inference, in outline

What it is. The mathematical engine. Exact Bayesian belief updating is intractable for any interesting model, so one optimizes an approximation: pick a family of manageable distributions Q , and adjust Q to minimize free energy $F = E_Q[\log Q(s) - \log P(o, s)]$, which equals the true surprise $-\log P(o)$ plus the divergence of Q from the exact posterior. Minimizing F therefore does two jobs at once — it scores the model and it *is* the belief update. In the driver model the "family" is the particle set: weighting and resampling are the minimization.

Why the main text skipped it. The particle filter can be understood operationally (chapter 06) without the variational story; the identity above adds rigor, not intuition.

Where to go. Any modern treatment of variational inference (Blei, Kucukelbir & McAuliffe, "Variational inference: a review for statisticians", JASA 2017). For the active-inference-specific assembly, the tutorial paper of Smith, Friston & Whyte, "A step-by-step tutorial on active inference" (Journal of Mathematical Psychology, 2022) — the standard entry point; discrete-state, see section E.

C. Markov blankets

What it is. The formal device separating "a thing" from "its environment": a set of states (sensory + active) through which all statistical influence between inside and outside must pass. The free-energy principle's broadest claims are stated in terms of blankets — a system's inside models its outside *because* the blanket makes their coupling indirect.

Why the main text skipped it. For a driver model, the blanket is trivial: the observation vector in, the control vector out. The concept earns its complexity only when one asks what counts as a system at all — a philosophy-of-science question.

Where to go. Kirchhoff et al., "The Markov blankets of life" (Journal of the Royal Society Interface, 2018) for the enthusiastic case; Bruineberg et al., "The Emperor's new Markov blankets" (Behavioral and Brain Sciences, 2022) for the sharpest critique — read together, they are the debate in miniature.

D. The biology and philosophy debate, with references

Chapter 01 gave the three-claim structure (process theory / universal principle / engineering framework) and took a position only on the third. The fuller reading list, both directions:

- **For the process theory:** Friston (2010) above; Parr, Pezzulo & Friston, *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior* (MIT Press, 2022) — the book-length statement, and the best single reference if a group buys one.
- **Sympathetic but independent:** Clark, "Whatever next? Predictive brains, situated agents, and the future of cognitive science" (Behavioral and Brain Sciences, 2013) — predictive processing without commitment to the strongest principle; the natural companion to *Great expectations* (Engström et al., 2018).
- **The critiques:** Bruineberg et al. (2022) above on blankets; Colombo & Wright, "First principles in the life sciences: the free-energy principle, organicism, and mechanism" (Synthese, 2021) on what the principle explains; the widely cited unfalsifiability worry is stated crisply in commentaries accompanying Clark (2013) and the BBS treatment of the principle.
- **The dark-room debate** (chapter 12's FAQ): Friston, Thornton & Clark, "Free-energy minimization and the dark-room problem" (Frontiers in Psychology, 2012).

Our position, restated for the record: the driver model's evidential standing rests on held-out prediction and ablation [Paper] [OSF], and would be unchanged by any outcome of this debate [Opinion].

E. The discrete-state formulation, and pymdp

What it is. Most of the active-inference literature — including nearly all tutorials — works with small discrete state spaces: a handful of states, observations, and actions, with the generative model written as labeled probability matrices (A for observation likelihoods, B for transitions, C for preferences, D for initial beliefs). Expected free energy is then a sum over a few dozen entries, and everything can be inspected by hand. pymdp (Heins et al., Journal of Open Source Software, 2022) is the reference Python library for exactly this.

Why the main text skipped it. The driving problem is continuous, high-dimensional, and long-horizon; none of the matrix machinery transfers directly, and the Waymo/TU Delft line had to build the particle-filter/CEM architecture *because* the tutorial formulation does not scale to it. A reader who learns the discrete formulation first must then unlearn the expectation that A, B, C, D matrices will appear in this codebase — they never do.

Why it is still worth a day of someone's time. The discrete world is where the *concepts* — preference as probability, epistemic value as expected information gain, the pragmatic/epistemic split — can be verified by hand on paper, which some readers find is what finally makes them click. The pymdp tutorial notebook ("active inference from scratch") is the recommended single exercise; its risk/ambiguity decomposition of expected free energy is an equivalent split to the pragmatic/epistemic one used here, and mapping one onto the other is a genuinely instructive afternoon.

F. Where the equations of this specific model live

For the reader who wants the real thing rather than any tutorial: the Supplementary Information of Schumann et al. (2026) is self-contained and, in our experience, more readable than the framework literature [Opinion]. The map: §2.2 observation model and looming; §2.3 beliefs; §2.4 preferences (Eqs. 44–52); §2.5 planning and evidence accumulation.